

Partial Notes for Statistical Thermodynamics (CHEM365)

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Chapter 1

A Brief Introduction

1.1 Motivation

This class is an intensive introduction to an important class of theoretical and computational methods used in modern science: the methods of statistical mechanics. Our goal is to introduce the theory behind what is empirically taught as thermodynamics and to build basic skills in computational and theoretical analysis. The focus in this class is on chemistry - the discipline where much of the development of statistical mechanics historically has taken place, but as we progress we will pay special attention to surprising parallels with biology, physics, and computer science demonstrating the breadth of impact of statistical mechanics.

Our goal is to learn the basics, so we will carefully learn the traditional topics: ensemble theory, simple magnetic systems, theory of polymers, and the basics of molecular dynamics simulations. This material tends to be somewhat dry, formal, and to those who do not know (yet) how to 'breath life' into equations, it may be somewhat difficult. Our hope is that the breadth of impact of statistical mechanics will compensate fully for the effort to learn it.

I would like to give you a quick summary of what we will discuss in this class intentionally made a bit surrealistic to hopefully awaken your curiosity: in second year physical chemistry classes concepts like temperature and chemical potential are introduced. They are introduced but they are taken for granted - if we try to think deeply about them, they do not make too much sense. What are they really? Quite surprisingly, we will see in this class that it turns out that they arise naturally when we simply try to generate maximally unbiased models of the world. This will perhaps help clarify the basic ideas behind thermodynamics that without statistical mechanics we simply accept without understanding, or in the very least this will provide an additional point of view. Furthermore, we will take a look at what models of magnets have in common with models of neurons in the brain and learn about basic behaviors of spin systems, for example, we will study magnetization and magnetic phase transitions. We will then combine what we have learned about maximum entropy unbiased modeling with what we have learned about magnets and find out why models of magnets are used to study protein folding and chromatin architecture (among other things).

Finally, we will see that we can learn about behaviors of complex molecules from very simple models - the kind of models that can be solved with pen and paper. Of course, powerful computers are needed if we wish to model molecules in more sophisticated ways.

We will see how this is done and what we learn from simple simulations. Long story short, by the end of this class you will find out what magnets, neurons, temperature, and proteins all have in common, what we can learn about nature's most important polymers by pretending they are just beads on a string, and what it takes to model them on a computer.

1.2 Organizational

This collection of notes is based on the CHEM365 class that I taught in the Winter 2021 semester at McGill University. These notes are very minimal and will be developed further in course of this semester. They should not be used as the sole source of content for this class, but they will assist us as we develop the material during live lectures. I will update these notes as we go by filling in some new material, adding explanations, corrections, etc..

These notes at the moment contain five topics:

1. Review of thermodynamics
2. Ensemble theory
3. Spin systems
4. Polymers
5. Computational methods: learning from data and from simulations.

The first three of these modules are covered in these notes almost fully. The fourth one is missing. Module 5 is covered in part and it will become your final project. The emphasis is placed on developing a deep understanding and control over the concepts and the math that give rise to the extraordinarily useful methods of thermodynamics and statistical mechanics when applied to problems in Chemistry, Biochemistry, and Physics.

One of the objectives of this class is to explore how thermodynamics emerges from first principles by means of statistical mechanics, and how it is used in modern research to study complex molecular systems. We will derive the Maxwell and Boltzmann distributions, find out the mathematical meaning of Temperature and Entropy, explore the relationship between Heat Capacity and our system's energy fluctuations, and learn how to use simulations to study molecular systems. The second objective is to introduce spin models and explore how they can be used to infer models of physical systems directly from experimental data. Time-permitting, polymers will be discussed as well.

In Chapters/Lectures 2-4 we will discuss some of the key points of thermodynamics with a particular focus on the Second Law and the Carnot engine. This discussion is meant as a review or a reminder of basic thermodynamics that are typically covered in second year physical chemistry or physics courses, and it will set the stage for the discussion of statistical mechanics in later chapters.

Chapter 2

Thermodynamics

2.1 Review of Basic Thermodynamics

Thermodynamics as a discipline is concerned with energy inter-conversion processes: heat is being converted into work throughout the world starting from industrial engines and ending with proteins in your cells. From the first law we know that energy can be converted into heat and into work, the second law tells us that a process of inter-conversion is spontaneous only if the entropy of the universe increases. Third law clarifies what vanishing entropy means. The rest is commentary.

Since classical thermodynamics is actually not the main focus of this class, our goal in this review of thermodynamics is to refresh some of the basic concepts, to present them more formally and more rigorously, perhaps at the expense of intuition. The expectation is that your intuitive understanding of thermodynamics was already developed in previous classes. We will discuss the laws of thermodynamics, work through basic energy conversion scenarios and phenomena, and discuss the Carnot engine. We will then use the Carnot engine to discuss in more depth the fundamental laws of thermodynamics completing a circle. Our working assumption is that this is not the first time you see this material and so some parts are presented with the assumption of familiarity with this material on part of the reader.

Let's recall some basics: the four laws of thermodynamics:

1. The Zeroth Law (the working principle of a thermometer!): two bodies in equilibrium with each-other have the same temperatures.
2. The First Law: The energy of a closed system (a universe) cannot be created or destroyed but it can be separated into two forms

$$dU = dQ + dW$$

heat (Q) and work (W), and it may be re-distributed and transferred between them (heat can be converted into work and vice versa).

3. The Second Law: Entropy

$$dS = \frac{dQ_{rev}}{T}$$

is a state function. Spontaneous processes increase or keep the same the Entropy in a closed system (the universe).

4. The Third Law: The entropy of a perfect crystal at $T=0$ is zero.

It may be difficult to see what these laws amount to just by reading them. For now an interesting thing to note is that quantum mechanics restricts how low we may hope to reduce the temperature of a perfect crystal, so the temperature can never actually be reduced to the absolute zero, not even in principle. We will see soon, this means that heat can never be transformed into work perfectly.

2.1.1 An archetypal scenario: expansion/compression of gas

Let's take a look at a particular system: gas confined in a container. In this type of experiments the volume, the temperature, and the pressure in the container may be controlled. The temperature is maintained by an external heat reservoir sometimes referred to as a heat bath. If we keep one of the variables fixed, for example if we keep the temperature fixed, any process (compression/expansion) will take place in a 2D space of the remaining two variables, in this case volume and pressure and P-V diagrams are often used to depict trajectories of gas systems.

One of key goals of thermodynamic analysis is to keep track of transformations of energy as our system moves from one state to another in the T-P-V space. To do this we use the First law of Thermodynamics, which states that the total energy in an isolated system (we can pretend an isolated system is its own universe) is conserved and may not be destroyed or created. Energy (U) can, however, be converted from heat to work and from work (w or W) to heat (q or Q). The mathematical statement of this fact is given by:

$$dU = dq + dw = TdS - P_{ext}dV \quad (2.1)$$

where by Second Law we have $dq = dq_{rev} = TdS$ while $dw = -P_{ext}dV$. By observing the differential dU we can clearly see that internal energy U depends on S : entropy and V : volume.

Ideal and non-ideal gas

The ideal gas equation is given by

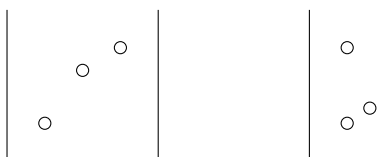
$$PV = nRT \quad (2.2)$$

where P is the pressure, V is the volume, T is temperature, n is the number of mols and R is the gas constant.

In contrast, real gases tend to behave in non-ideal ways. van der Waals equation of state is often used to describe the behavior of a real gas with excluded volume and interactions:

$$P = \frac{nRT}{V - nb} - \frac{an^2}{v^2} \quad (2.3)$$

where a and b are constants. Let's take a look at an isothermal compression of the van der Waals gas and find the change in internal energy due to change in volume. In other words we are looking for $\Delta U|_{V_1}^{V_2}$.



The diagram above represents the initial and final states of our system.

Now, I'll demonstrate how to approach any problem that looks unfamiliar or the solution to which you do not immediately know. The trick is to apply the Scientific Method. Contrary to common belief the scientific method does not only belong in a lab, it can be applied to any problem, certainly to any problem in this class, but even more broadly to something like... motorcycle maintenance! So, here we will break the problem of finding ΔU into smaller sub-problems and then apply the algorithm: Step 1. Make an educated guess, i.e., form a hypothesis. Step 2. Test the hypothesis; 3. Conclude and generate the next hypothesis. Step 4. Repeat steps 1-3 until all uncertainty evaporates. To experienced students this may sound like an overkill and for many problems it clearly is. But sooner or later you will likely encounter a problem that will be somewhat stubborn, and I would like to make sure that you have the tools you need to deal with this kind of difficulties. So, let's do an example.

So, we wish to compute the change in internal energy of a gas following a compression. Where do we start? Let's ask a simple question:

Sub-problem 1: What is ΔU ?

Hypothesis

Test

Conclusion

Sub-problem 2: What is dU in terms of what we know? We know the equation of state and the fact that the temperature is constant.

Hypothesis $dU = TdS - PdV$

Test $dU = dq + dw = TdS - PdV$

Conclusion OK. But what do I do with dS ?

Sub-problem 3 Since our goal is to integrate dU over expansion in volume ideally we need an expression that only contains a volume differential dV . The problematic part is dS , we need to re-express this in terms of Volume and Temperature

Hypothesis If I express dS as a differential in V and T , I will be able to write dU in terms of volume.

Test $dS = \frac{\partial S}{\partial V}dV + \frac{\partial S}{\partial T}dT$, so for an isothermal process $dT=0$, $dS = \frac{\partial S}{\partial V}dV$.

Conclusion We replace dS in dU with the above and we now only integrate with respect to volume, great! But we still do not know how entropy S depends on volume V , so we cannot take the integral yet.

Sub-problem 4 Equations of state give us relationships between P , V and T , they don't tell us anything about the entropy S . To make progress we need to replace S with some function of P , V , and/or T .

Hypothesis Maxwell relations are sometimes helpful with variable changes, let's see if we can find anything useful.

Test We may use this relation: $\frac{\partial S}{\partial V} = \left(\frac{\partial P}{\partial T}\right)_V$ and get our final expression

Conclusion $dU = T\left(\frac{\partial P}{\partial T}\right)_V dV - PdV$

At this point we simply choose the kind of gas we wish to use in our problem, and if our process is done reversibly - we use its equation of state to express pressure and integrate this expression to obtain the answer:

$$\Delta U = T \int_{V_1}^{V_2} \left(\frac{\partial P}{\partial T}\right)_V dV - \int_{V_1}^{V_2} PdV \quad (2.4)$$

2.2 New Thermodynamic State Functions via the Legendre Transform

As we have said before, the object of thermodynamics is to follow free energy inter-conversion during various processes. The internal energy $U(S, V)$ is a function of entropy and volume and it is often impractical to try to measure or calculate because Entropy (S) is an inconvenient variable; Convenient variables are those that can be easily become 'knobs' that can be turned in experiments, but it is hard to control the entropy in a lab. Comparatively, Pressure and Temperature are a lot easier to tune in a lab. Luckily, there is a mathematical

way to switch from one variable to another, it is called the Legendre transform. By applying Legendre transform to internal energy U we may obtain functions that contain the same information but depend on more convenient variables. Examples of such functions are the familiar Enthalpy, Helmholtz free energy, and Gibbs free energy. You have certainly encountered these functions in the past.

We will define the Legendre transform first in general term for some function f that depends on two independent variables x and y .

$$g_1(r, y) = f(x, y) - rx \quad (2.5)$$

$$g_2(x, t) = f(x, y) - ty \quad (2.6)$$

$$g_3(r, t) = f(x, y) - rx - ty \quad (2.7)$$

where the new variables $r = \frac{\partial f}{\partial x}$, and $t = \frac{\partial f}{\partial y}$ are called conjugate variables to x and y respectively.

Let's see how we can apply the Legendre transform to generate new thermodynamics functions starting from the internal energy $U(S, V)$:

$$U(S, V) \rightarrow H(S, P) = U - (-P)V = U + PV \quad (\text{Enthalpy})$$

$$U(S, V) \rightarrow A(T, V) = U - TS \quad (\text{Helmholtz free energy})$$

$$U(S, V) \rightarrow G(T, P) = U + PV - TS \quad (\text{Gibbs free energy})$$

In order to see more clearly that the new functions depend on new variables we will use the differential form of internal energy $dU = TdS - PdV$ and write out the differentials of the new thermodynamic functions. Starting with Enthalpy we get: $dH = dU + d(PV) = dU + VdP + PdV = TdS - PdV + VdP + PdV = TdS + VdP$ which clearly shows that enthalpy 'lives' in the space of entropy S and pressure P (these are the variable that are allowed to change). For Helmholtz free energy we get $dA = dU - d(TS) = TdS - PdV - TdS - SdT = -PdV - SdT$ which shows that A is an energy function in the space of volume and temperature. Repeating this for the Gibbs free energy is left as an exercise to the reader.

2.3 Maxwell Relations

Maxwell relations are nothing but consequences of mathematical properties of state functions:

$$\frac{\partial}{\partial y} \left(\frac{\partial f(x, y)}{\partial x} \right) = \frac{\partial}{\partial x} \left(\frac{\partial f(x, y)}{\partial y} \right).$$

$$dU = TdS - PdV \text{ and } \left(\frac{\partial U}{\partial S} \right)_V = T \text{ and } \left(\frac{\partial U}{\partial V} \right)_S = -P \text{ so we have}$$

$$\frac{\partial}{\partial y} \left(\frac{\partial f(x,y)}{\partial x} \right) = \frac{\partial}{\partial x} \left(\frac{\partial f(x,y)}{\partial y} \right) \Rightarrow \left(\frac{\partial T}{\partial V} \right)_S = - \left(\frac{\partial P}{\partial S} \right)_V$$

We can do the same with Enthalpy and the Free Energies and obtain the other Maxwell relations:

$$dH = TdS + VdP, \left(\frac{\partial H}{\partial S} \right)_P = T, \left(\frac{\partial H}{\partial P} \right)_S = V \\ \Rightarrow \left(\frac{\partial T}{\partial P} \right)_S = \left(\frac{\partial V}{\partial S} \right)_P$$

$$dA = -SdT - PdV, \left(\frac{\partial A}{\partial T} \right)_V = -S, \left(\frac{\partial A}{\partial V} \right)_T = P \\ \Rightarrow \left(\frac{\partial S}{\partial V} \right)_T = \left(\frac{\partial P}{\partial T} \right)_V$$

Also note the inverse relation:

$$\frac{\partial f}{\partial x} = \frac{1}{\left(\frac{\partial x}{\partial f} \right)} \Rightarrow \text{using the above for } \left(\frac{\partial T}{\partial V} \right)_S \rightarrow \left(\frac{\partial V}{\partial T} \right)_S = - \left(\frac{\partial S}{\partial P} \right)_V$$

Chapter 3

Thermodynamic Processes

3.1 Joule-Thompson Experiment/Coefficient

Joule-Thompson Experiment dates back to 1852. It is named after James Prescott Joule and William Thomson (Lord Kelvin), two brilliant physicists who made many fundamental discoveries. The idea in this experiment is to observe the consequences of the equipartition of energy by tracking the temperature of a gas that is allowed to expand adiabatically from volume V_1 , pressure P_1 to volume V_2 and pressure P_2 , with $P_2 < P_1$ and $V_2 > V_1$. This (or similar) process is used in refrigeration and for liquefaction of gases (e.g. Hampson–Linde cycle).

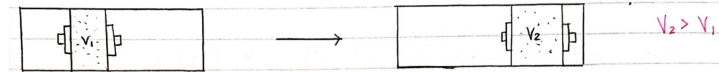


Figure 3.1: Joule-Thompson Experiment

$$W = W_{leftpiston} + W_{rightpiston} = -\int_{V_1}^0 P_1 dV - \int_0^{V_2} P_2 dV = P_1 V_1 - P_2 V_2$$

$q = 0$ for an insulated system,

$$\Delta U = q + w = U_2 - U_1 = w = P_1 V_1 - P_2 V_2,$$

$$U_2 + P_2 V_2 = U_1 + P_1 V_1,$$

$$H_1 = H_2 \Rightarrow \Delta H = 0,$$

$$dH = C_p dT + \left(\frac{\partial H}{\partial P}\right)_T dP = 0 \text{ and dividing through by } dP,$$

$$C_p \frac{dT}{dP} + \left(\frac{\partial H}{\partial P}\right)_T = 0 \Rightarrow \left(\frac{\partial H}{\partial P}\right)_T = -C_p \mu_{J-T},$$

where μ_{J-T} is called the Joule-Thompson coefficient.

What is μ_{J-T} for an ideal gas?

$$\left(\frac{\partial H}{\partial P}\right)_T = \left[\frac{\partial U}{\partial V} + P\right] \frac{\partial V}{\partial P} + V, \quad \left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial P}{\partial T}\right)_V - P$$

Recall from (2.2) (ideal gas law) that $P = \frac{nRT}{V}$ so that $\left(\frac{\partial U}{\partial V}\right)_T = nRT/V - nRT/V = 0$ so $\left(\frac{\partial H}{\partial P}\right)_T = -\frac{nRT}{V} \left(\frac{V^2}{nRT}\right) + V = 0$ and thus

$$\mu_{J-T} = -\frac{1}{C_p} \left(\frac{\partial H}{\partial P}\right)_T = 0 \text{ (for ideal gas)}$$

So, interactions between particles are essential for the cooling or heating effects that may arise. We will explore this in the home work assignment.

A note: In general, the internal energy of the system is given by an energy function, a Hamiltonian H which represents the total energy of the system and is given by

$$H = T(p_1, p_2, p_3, \dots) + V(x_1, x_2, x_3, \dots) \quad (3.1)$$

where T is the total kinetic energy summing over all particles, $\sum_i \frac{p_i^2}{2m_i}$, for particles with masses m_i and $V(x_1, x_2, \dots)$ captures the potential energy term, i.e., the interaction between two or more particles that depends on their distance from each other, but not on momenta.

3.2 Ideal Gas and Quasi-Static Adiabatic Processes

The equation of state for an ideal gas is given by Eqn. (2.2) (ideal gas law), i.e.,

$$PV = nRT$$

Let's recall that

$$dU = dQ + dw = dQ - PdV \text{ and } dQ = dU + PdV, \text{ and that}$$

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = \frac{dU}{dT} \Rightarrow dU = C_V dT \text{ (for an ideal gas).}$$

Taking the differential of the equation of state for the ideal gas :

$$d(PV) = d(nRT) \Rightarrow dPV + VdP = nRdT \text{ (by the product rule).}$$

$$\text{Then } dQ = dU + PdV = C_V dT + nRdT - VdP = (C_V + nR)dT - VdP$$

$$\text{and assuming constant pressure, } dQ_P = (C_V + nR)dT \Rightarrow C_p = C_V + nR$$

Quasi-Static Adiabatic Process

A quasi-static process is a processes that is performed sufficiently slowly in a way that the equilibrium between the system and the surroundings is maintained at all times. In this situation the equation of state can be used to describe the system as it undergoes the process

in question. We will use ideal gas as a toy model to explore what such processes entail but a similar line of reasoning translates to molecular and Nano-scale systems that are of chemical interest today. For molecular system we talk about the concept of separation of time-scales which we will likely discuss later in this class.

For an adiabatic process $dQ = 0$.

$$dQ = C_V dT + nR dT - V dP \Rightarrow dQ = (C_V + nR) dT - V dP,$$

$$C_P dT = V dP \Rightarrow dT = \frac{V dP}{C_P}$$

$$C_V dT = -P dV \Rightarrow dT = -\frac{P dV}{C_V}$$

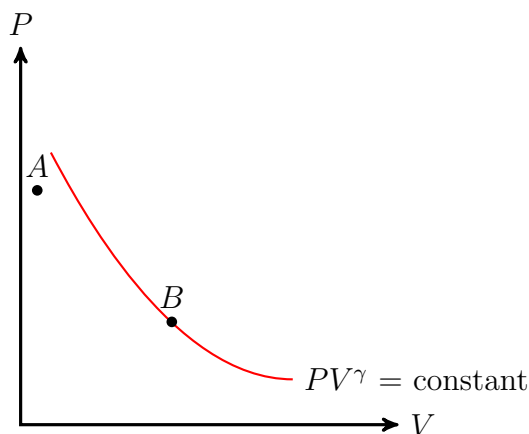
Which means $\frac{V dP}{C_P} = -\frac{P dV}{C_V}$ Rearranging and integrating both sides,

$$\int \frac{dP}{P} = -\int \frac{C_P}{C_V} \frac{dV}{V} \rightarrow \int \frac{dP}{P} = -\gamma \int \frac{dV}{V},$$

where we have let $\gamma = \frac{C_P}{C_V}$. The outcome of this integration is

$$\ln\left(\frac{P_B}{P_A}\right) = -\gamma \ln\left(\frac{V_B}{V_A}\right) = \gamma \ln\left(\frac{V_A}{V_B}\right).$$

Taking exponential on both sides leads to $P_B(V_B)^\gamma = P_A(V_A)^\gamma$. This tells us that PV^γ is fixed in an adiabatic quasi-static process. This result will be helpful to us.



A quick summary: So far we have generated thermodynamic potentials starting from the internal energy using the Legendre transform and we have derived useful relations (e.g., Maxwell relations). We have explored at isothermal and adiabatic processes for ideal and non-ideal gases; Next, we will combine these two processes in a cycle in order to construct an idealized engine, the archetype of every engine in existence, the Carnot engine and use it, following the founders of the field of thermodynamics, in order to understand deeper the Second Law of Thermodynamics.

Chapter 4

Carnot Engines and The Second Law

4.1 The Carnot Engine

Carnot Engine - Ideal Gas, Reversible Process

These notes use the convention $\Delta U = Q + W$, where W is the work that is being done by the system and it is negative when the system is expanding (step AB and BC) and positive when it is contracting (steps CD and DA). *Engine:*

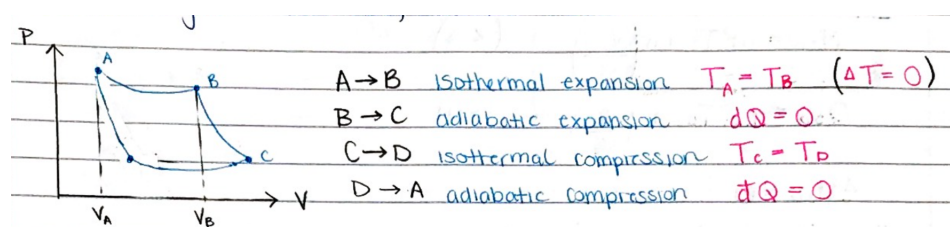


Figure 4.1: Carnot cycle for reversible perfect gas.

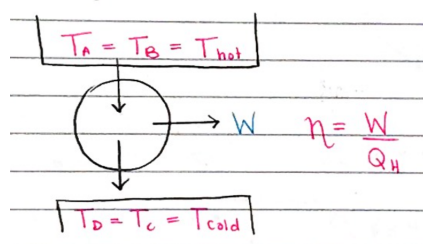


Figure 4.2: Carnot heat engine diagram

$A \rightarrow B$:

$$W_{AB} = - \int_A^B P dV = -nRT \int_A^B \frac{dV}{V} = -nRT \ln\left(\frac{V_B}{V_A}\right) < 0$$

so that

$$Q_{AB} = -W_{AB} = nRT_A \ln\left(\frac{V_B}{V_A}\right)$$

$B \rightarrow C$:

$$PV^\gamma = P_B V_B^\gamma = P_C V_C^\gamma$$

$$W_{BC} = - \int_B^C P dV = -P_B V_B^\gamma \int_B^C \frac{dV}{V^\gamma} = -\frac{1}{1-\gamma} P_B V_B^\gamma V^{-(\gamma-1)} \Big|_B^C$$

which is just

$$\Rightarrow -\frac{1}{1-\gamma} P_C V_C + \frac{1}{1-\gamma} P_B V_B = \frac{1}{1-\gamma} (P_B V_B - P_C V_C) < 0$$

It is easy to see if we recall that $P_B V_B = nRT_B$ and $P_C V_C = nRT_C$, and $T_C < T_B \Rightarrow P_B V_B - P_C V_C > 0$ while $1 - \gamma < 0$ because $\gamma = \frac{C_P}{C_V}$ and $C_P > C_V$.

$C \rightarrow D$:

$$W_{CD} = -nRT_C \ln\left(\frac{V_D}{V_C}\right) > 0$$

$$Q_{CD} = nRT_C \ln\left(\frac{V_D}{V_C}\right)$$

$$\Delta U_{CD} = 0$$

$D \rightarrow A$:

$$W_{DA} = -\frac{1}{1-\gamma} (P_A V_A - P_D V_D) > 0$$

$$Q_{DA} = 0$$

$$\Delta U_{AB} = W_{DA}$$

Total Work:

$$W_{tot} = -nRT_A \ln\left(\frac{V_B}{V_A}\right) - nRT_C \ln\left(\frac{V_D}{V_C}\right) + \frac{1}{1-\gamma} (P_B V_B - P_C V_C + P_D V_D - P_A V_A)$$

We simplify this using the ideal gas equation of state $PV = nRT$

$$W_{tot} = -nRT_A \ln\left(\frac{V_B}{V_A}\right) - nRT_C \ln\left(\frac{V_D}{V_C}\right) - \frac{nR}{1-\gamma} (T_B + T_D - T_C - T_A)$$

$$P_B V_B^\gamma = P_C V_C^\gamma \quad P_B V_B = nRT_B \rightarrow nRT_B V_B^{\gamma-1} = nRT_C V_C^{\gamma-1}$$

$$P_A V_A^\gamma = P_D V_D^\gamma \quad P_A V_A = nRT_A \rightarrow nRT_A V_A^{\gamma-1} = nRT_D V_D^{\gamma-1}$$

$$\frac{T_B}{T_C} = \left(\frac{V_C}{V_B}\right)^{\gamma-1}; \quad \frac{T_A}{T_D} = \left(\frac{V_D}{V_A}\right)^{\gamma-1}; \quad T_A = T_B \text{ and } T_C = T_D$$

$$\Rightarrow \frac{V_C}{V_B} = \frac{V_D}{V_A} \Rightarrow \frac{V_C}{V_D} = \frac{V_B}{V_A}$$

Remember that the **efficiency** is given by

$$\eta = \left| \frac{W}{Q} \right|$$

so that after cancellations we get

$$\eta = \left| \frac{-nRT_A \ln\left(\frac{V_B}{V_A}\right) - nRT_C \ln\left(\frac{V_D}{V_C}\right)}{nRT_A \ln\left(\frac{V_B}{V_A}\right)} \right| = \frac{nRT_A \ln\left(\frac{V_D}{V_C}\right) - nRT_C \ln\left(\frac{V_D}{V_C}\right)}{nRT_A \ln\left(\frac{V_D}{V_C}\right)} = \frac{T_A - T_C}{T_A}$$

In other words, we arrive at the famous result, the efficiency of the Carnot engine

$$\eta = 1 - \frac{T_C}{T_A}$$

which tells us a remarkable thing: the efficiency of the Carnot engine is independent of the details of the engine, independent of the properties of the gas, it only depends on the temperatures of heat reservoirs between which the engine operates. We will see in the next section that this result helped formulate some of the most basic laws in Science, the Second Law of Thermodynamics.

4.2 The Second Law of Thermodynamics

Kelvin-Planck Statement of the 2nd Law

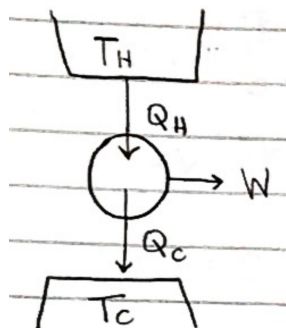


Figure 4.3: Diagram for Kelvin-Planck 2nd Law statement.

This version of the 2nd law states that it is impossible to build an engine with $Q_C = 0$

This is a good point to note that the 3rd law of thermodynamics sets the main constraint on the efficiency of the Carnot ideal engine: T_C can never be equal to 0.

Clausius Statement of the 2nd Law

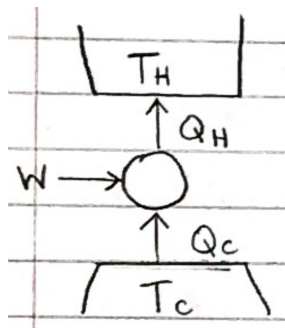


Figure 4.4: Diagram for Clausius 2nd Law statement.

This version of the 2nd law states that it is impossible to build a refrigerator with $W = 0$, see the diagram in Fig.4.4.

Proof that these statements are equivalent: it is sufficient to demonstrate that if we assume that the Clausius statement is false, the Kelvin-Plank statement must be false as well and vice versa, see Fig. 4.5.

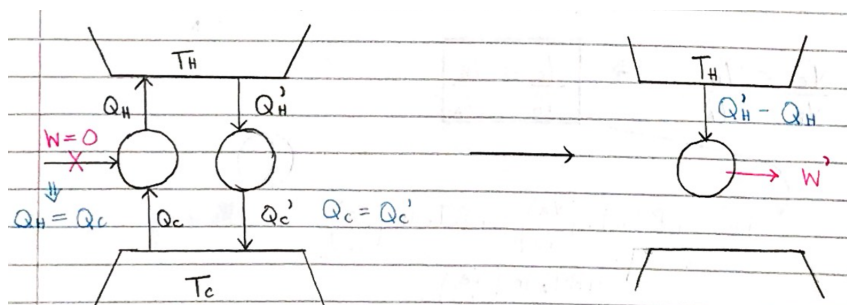


Figure 4.5: Proof of the equivalence of the Kelvin-Plank and the Clausius statements.

Carnot Statement:

'No engine can be more efficient than the Carnot engine.'

We will prove that contradiction of this statement contradicts Clausius statement of the Second Law and thereby we will establish the equivalence between the Carnot statement and the Clausius statement. We will start by assuming that an engine that is more efficient than Carnot is possible. The efficiency of this engine is greater than the efficiency of the Carnot engine: $\eta > \eta_{CA}$, where η is the efficiency of the hypothetical engine and η_{CA} refers

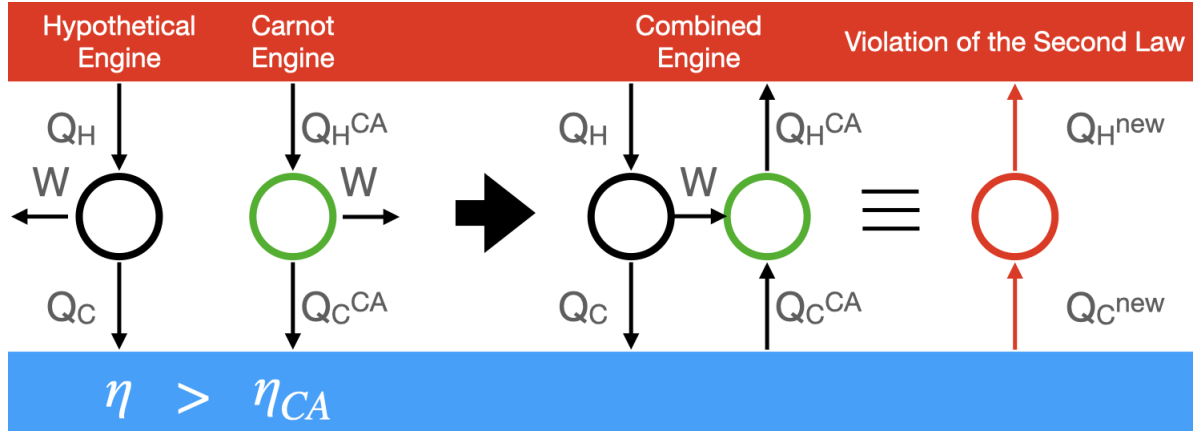


Figure 4.6: A pictorial demonstration by contradiction that the Carnot statement is equivalent to the Clausius statement of the Second Law

to the efficiency of the Carnot engine. First, we infer that

$$\eta > \eta_{CA} \implies \frac{W}{Q_H} > \frac{W_{CA}}{Q_H^{CA}}$$

where all we have done is used the definition of efficiency, $\eta = \frac{\text{work}}{\text{input heat}}$. Without loss of generality we let $W = W_{CA}$ and we cross multiply to obtain the statement

$$Q_H^{CA} > Q_H$$

and immediately see that

$$Q_C^{CA} > Q_C$$

as well. Now we will construct a new engine in which we allow the work from the hypothetical more efficient engine to power the Carnot engine running in reverse. As a result, we will get a new engine that exchanges $Q_H^{new} = Q_H - Q_H^{CA} < 0$ with the hot reservoir, produces no net work, and exchanges $Q_C^{new} = Q_C - Q_C^{CA} < 0$ with the cold reservoir. Negative sign in Q_H^{new} means that the heat now flows in the direction opposite to Q_H (which in this new construction flows from heat reservoir into the system), and the negative sign in Q_C^{new} implies that the heat exchanged with the cold reservoirs similarly flows in the opposite direction from Q_C , i.e., Q_C^{new} runs from the cold reservoir into the system. This amounts to pumping heat from cold reservoir to hot reservoir without the investment of work in a direct contradiction of the Clausius statement of the Second Law. This implies that the Second Law prohibits the construction of engines more efficient than the Carnot engine.

We finish our discussion of thermodynamics with the derivation of the Clausius inequality starting from the Carnot statement of the second law. Before we proceed we will introduce a new state function - the Entropy defined as $S = \frac{Q_{rev}}{T}$. It was discovered empirically and named by Clausius. The fact that Entropy is a state function implies that its integral over a closed path vanishes - a property that we will use to restate the Second Law in terms of the Entropy of the universe.

The Carnot engine operates reversibly between two heat reservoirs held at distinct temperatures T_H and T_C . A non-Carnot engine would involve some irreversibility and we will

use only this feature to develop our argument. For an irreversible process,

$$\eta_{irr} = \frac{|W|}{Q_H} = \frac{Q_H - |Q_C|}{Q_H} = 1 - \frac{|Q_C|}{Q_H} < 1 - \frac{T_C}{T_H}$$

so that

$$\frac{Q_H}{T_H} - \frac{|Q_C|}{T_C} < 0$$

and using the paths in the Carnot cycle we infer from the above that

$$\oint \frac{dQ_{irr}}{T} < 0 = \oint \frac{dQ_{rev}}{T} = \oint dS$$

.

$$\frac{dQ_{irr}}{T} - \frac{dQ_{rev}}{T} = \frac{dQ_{irr}}{T} - dS < 0$$

so that

$$dS \geq \frac{dQ}{T}$$

known as the Clausius inequality.

Finally, we may use this inequality to get the better known formula for the Second Law which summarizes arguably the most important physical law as the entropy of a closed system ('the Universe') increases in spontaneous processes. The entropy of 'the Universe' is given by the entropies of the system and the surroundings combined:

$$dS_{univ} = dS_{surr} + dS_{sys}.$$

For engines we are interested in a cyclic paths:

$$\oint dS_{univ} = \oint dS_{surr} + \oint dS_{sys}$$

where

$$\oint dS_{surr} = - \oint \frac{Q_{irr}}{T} \geq 0$$

$$\oint dS = 0,$$

and as a consequence

$$\oint dS_{univ} \geq 0.$$

Chapter 5

Pre-Ensemble Theory

5.1 Probability

-Discrete Probability:

$x \in \{1, 2, 3, \dots, m\}$, associated probabilities are $P(1), P(2), \dots, P(m)$ and $\sum_{i=1}^m P(i) = 1$

-Continuous Probability:

$a \leq x \leq b$ for real numbers a, b and $\int P(x)dx = 1$.

Note:

$P(i \text{ or } j) = P(i) + P(j)$ or $P(i, j) = P(i)P(j)$ if they are uncorrelated.

-Marginal probability

$P_1(i) = \sum_j P(i, j)$ (w.r.t the 1st variable)

$P_2(i) = \sum_j P(i, j)$ (w.r.t the 2nd variable)

If uncorrelated: $P(i, j) = P(i)P(j)$,

$P_1(i) = \sum_j P(i)P(j) = P(i) \sum_j P(j) = P(i)(1) = P(i)$ (recall discrete case).

-Conditional Probability

$P(i|j) = \frac{P(i,j)}{P_2(j)}$

What do we do if we know a distribution ? We calculate observables - averages, variances, etc. of events

Example. Russian Roulette

On average, how many attempts does it take to shoot yourself?

$P_{\text{shot}} = 1/6$ $P_{\text{survive}} = 5/6$

$P_{\text{die after 6 shots}} = (P_{\text{survive}})^5 P_{\text{shot}} = (5/6)^5 (1/6)$

$\langle n \rangle_{P(n)} = \sum_n n P(n) = \sum_n n (5/6)^{n-1} (1/6) = \frac{1}{6} \sum_{n=1}^{\infty} n (5/6)^{n-1}$ `sp.Sum(n**(5.6)**(n-1),(n,1,sp.oo))`

=36

-Objective Probabilities

Frequency probability: $P(x) \sum_{i=1}^N \delta_{x_i, x}$ where $\delta_{a,b} = \begin{cases} 1 & \text{if } a = b \\ 0 & \text{if } a \neq b \end{cases}$, and is known as the 'Kronecker delta'.

Interpretation: probability is the relative frequency "in the long run" of outcomes.

Law of Large Numbers:

$$\lim_{N \rightarrow \infty} \langle x \rangle_{N \text{ measurements}} = \langle x \rangle_{\text{True}}$$

Gambler's fallacy: nothing in probability theory is predictive in terms of trends. It is incorrect to think if you've been above the mean for N trials then you will be below the mean for the next N trials.

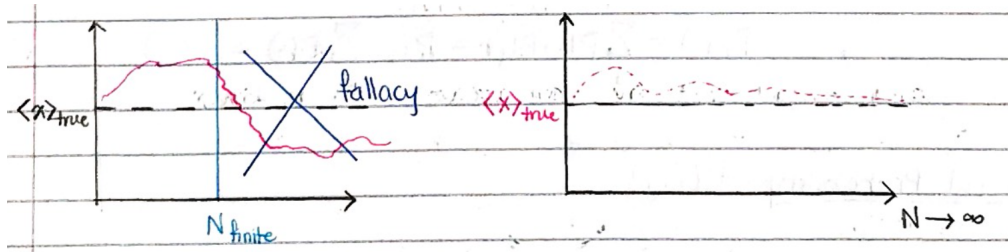


Figure 5.1: The Gambler's fallacy and some associated graphs.

-Subjective Probability *Probability:* degree of belief, a reasonable expectation representing a degree of knowledge

Bayes' Theorem: $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$ where $P(A|B)$ is posterior and $P(B|A)$, $P(A)$ and $P(B)$ are likelihood, prior and evidence knowledge, respectively.

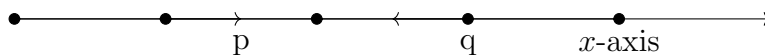
Example. $P(\text{fire}|\text{smoke}) = \frac{P(\text{smoke}|\text{fire})P(\text{fire})}{P(\text{smoke})}$

Base-rate fallacy

Example. Police test 1000 drivers at random for alcohol consumption. The device never misses a true drunk, however, it has a 5% false positive rate. 1 in 1000 drivers is really drunk. Suppose police stop a driver at random and test indicates they're drunk. How high is probability they are actually drunk?

Answer: $P(\text{drunk}|D) = \frac{P(D|\text{drunk})P(\text{drunk})}{P(D)}$ where D is a positive reading on the device.
 $P(\text{drunk}) = 0.001$; $P(D|\text{drunk}) = 1.0$; $P(\text{sober}) = 1.0 - 0.001 = 0.999$; $P(D|\text{sober}) = 0.05$
 $P(D) = P((D|\text{drunk})P(\text{drunk}) + P(D|\text{sober})P(\text{sober}) = 0.05095$
 $P(\text{drunk}|D) = \frac{(1)(0.001)}{0.05095} = 0.019627$ which is approximately 2%.

Random Walk



$q + p = 1$ and if no bias in either direction, $q = p = 1/2$

$x^N = \sum_{i=1}^N \sigma_i$ where x is the position after N steps and σ_i is $+1$ if you have moved right on a particular step i or -1 if you moved left.

$$P_\sigma = p\delta_{\sigma,+1} + q\delta_{\sigma,-1} = \text{where } \delta_{a,b} = \begin{cases} p & \text{if } \sigma = +1 \\ q & \text{if } \sigma = -1 \end{cases}$$

Probability for N steps:

$$P(\sigma_1, \dots, \sigma_N) = \prod_{j=1}^N P(\sigma_j)$$

Expectation Value (average) of x^n

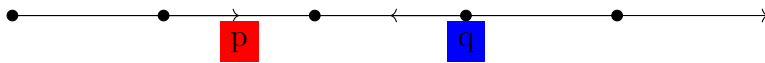
$$\langle x^N \rangle_{P(\sigma_1, \dots, \sigma_N)} = \left\langle \sum_{i=1}^N \sigma_i \right\rangle = \sum_{i=1}^N \langle \sigma_i \rangle = N \langle \sigma \rangle = N \sum_{\sigma=\pm 1} \sigma P(\sigma) = N(p - q) = N(2p - 1)$$

Example. Residual entropy of 1 mol of CO at 0 K:

$S = k_B \ln(\Omega)$ with 1 mol of 1 type of molecule and 2 configurations so $S = (6.022 \times 10^{23} k_B \ln(2)) \approx 5.76 \text{ J/K}$

Probability Homework Question

Random walk in 1-D



$$q + p = 1;$$

$x^N = \sum_{i=1}^N \sigma_i$ where x is the position after N steps and σ_i is $+1$ if you have moved right on a particular step i or -1 if you moved left.

$$\langle x^N \rangle_{P(\sigma_1, \dots, \sigma_N)} = \left\langle \sum_{i=1}^N \sigma_i \right\rangle = \sum_{i=1}^N \langle \sigma_i \rangle = N \langle \sigma \rangle = N \sum_{\sigma=\pm 1} \sigma P(\sigma) = N(p - q) = N(2p - 1)$$

$$\langle x^2 \rangle = \sum_N x^2 i P(x)$$

$$\langle x^2 \rangle = \sum_{j=1}^N \sum_{j'=1}^N \langle \sigma_j \sigma_{j'} \rangle = N^2(p - q)^2 + 4Npq$$

$$\sigma_j \sigma_{j'} = \delta_{j,j'} + (1 - \delta_{j,j'})(p - q)^2 = \begin{cases} 1, & j = j' \\ (p - q)^2, & j \neq j' \end{cases}$$

$$\text{Var}(x) = \langle (\Delta x)^2 \rangle = \langle (x - \langle x \rangle)^2 \rangle = \langle x^2 \rangle - \langle x \rangle^2 = 4Npq$$

$$\Delta x_{rms} = \sqrt{\text{Var}(x)} \text{ where (root mean square is rms)}$$

$P_{N,x} = \binom{N}{N_R} p^{N_R} q^{N_L}$ where N_R, N_L are number of steps right (R) or left (L) and $N = N_R + N_L$ and

$$\binom{N}{N_R} = \frac{N!}{N_R! N_L!} \text{ since } N_L = N - N_R$$

Chapter 6

Pre-ensemble Theory: Maxwell distribution of velocities in ideal gas

Introduction

Ideal gas is a collection of non-interacting particles confined to a volume. The thermodynamics state of ideal gas is completely characterized by temperature (T), pressure (P), volume (V), and number of particles(n) that are connected by the Ideal Gas Law

$$PV = nRT.$$

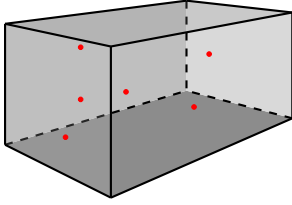


Illustration of ideal gas particles in a container.

Ideal gas particles are assumed to not interact with each other in any way. For this reason the energy describing this system contains only kinetic energy terms and no interaction terms are present:

$$E = \sum_{i=1}^N \frac{1}{2} m_i v_i^2 = \sum_{i=1}^N \frac{1}{2} m_i (v_{i,x}^2 + v_{i,y}^2 + v_{i,z}^2) \quad (6.1)$$

The absolute value of velocity $|v|$ is called speed and defined simply as

$$|v| = \sqrt{v_{i,x}^2 + v_{i,y}^2 + v_{i,z}^2} \quad (6.2)$$

The derivation of distribution of velocities in ideal gas

In this section we will infer the probability distribution of velocities of ideal gas particles from very few very basic assumptions. The probability we are interested in : $P(v_x, v_y, v_z)dv_x dv_y dv_z$

is the probability of observing a gas particle with velocity with components $v_x \in \{v_x, v_x + dv_x\}$, $v_y \in \{v_y, v_y + dv_y\}$, $v_z \in \{v_z, v_z + dv_z\}$

When this derivation was done, experimental techniques that could provide this information directly were not invented yet. Thinking about matter in terms of motions of individual molecules or atoms was a profoundly hypothetical and a theoretical exercise. A theoretical exercise that, as we know today, paid off in incredible ways. But within the contemporary context, the uncertainty about gases at that time was similar to the uncertainty we have today about white matter in the brain, non-coding genes in DNA, dark matter in the universe, evolution of viruses, etc... Following the steps Maxwell took in order to learn something very important about a system that was at that time inaccessible to direct exploration may inspire our own attempts to push the boundaries of what we know and understand. Learning a lot from very little is the name of the game, let's see how it is done.

Maxwell's Assumptions:

1. Components v_x, v_y, v_z are independent of one another so that $P(v_x, v_y, v_z) = g(v_x)g(v_y)g(v_z)$ (recall that the combined probability of several independent variable is just the product of their individual probabilities).
2. Space is isotropic (gas molecules do not prefer one direction over another). This amounts to the fact that $P(v_x, v_y, v_z)$ overall should depend only on the magnitude of velocity or the speed: $|v| = \sqrt{v_x^2 + v_y^2 + v_z^2}$, therefore we can write

$$P(v_x, v_y, v_z) = \Phi(|v|) = g(v_x)g(v_y)g(v_z).$$

Differentiating w.r.t v_x :

$$\frac{\partial \Phi(|v|)}{\partial v_x} = \frac{d\Phi}{d|v|} \frac{\partial |v|}{\partial v_x} = \frac{dg(v_x)}{dv_x} g(v_y)g(v_z),$$

and

$$\frac{\partial |v|}{\partial v_x} = \frac{\partial \sqrt{v_x^2 + v_y^2 + v_z^2}}{\partial v_x} = \frac{v_x}{|v|},$$

so that

$$\frac{d\Phi}{d|v|} \frac{v_x}{|v|} = \frac{dg(v_x)}{dv_x} g(v_y)g(v_z).$$

Moving all terms that depend on $|v|$ to the left-hand side, and all terms that depend on v_x to the right-hand side of the equation we get

$$\frac{d\Phi}{d|v|} \frac{1}{|v|} = \frac{1}{v_x} \frac{dg(v_x)}{dv_x} g(v_y)g(v_z) \Rightarrow \frac{d\Phi}{d|v|} \frac{1}{\Phi|v|} = \frac{dg(v_x)}{dv_x} \frac{1}{g(v_x)v_x}$$

where we have used the fact that $\frac{\Phi}{g(v_x)} = g(v_y)g(v_z)$ obtained by rearranging the expression in Assumption 1.

Repeating this procedure for v_y and v_z we obtain:

$$\frac{\Phi'}{\Phi|v|} = \frac{g'(v_x)}{g(v_x)v_x} = \frac{g'(v_y)}{g(v_y)v_y} = \frac{g'(v_z)}{g(v_z)v_z}.$$

We now notice that because v_x, v_y, v_z are independent, the equality $\frac{g'(v_x)}{g(v_x)v_x} = \frac{g'(v_y)}{g(v_y)v_y} = \frac{g'(v_z)}{g(v_z)v_z}$ implies that $\frac{g'}{g*v}$ does not depend on its argument at all (consider choosing for example $v_x = 10$, $v_u = 5$, and $v_z = 1001$ we get that the value of our function for one argument is equal to its value for any other argument), i.e., it's constant:

$$\frac{\Phi'(|v|)}{\Phi|v|} = \frac{g'(v_x)}{g(v_x)v_x} = \frac{g'(v_y)}{g(v_y)v_y} = \frac{g'(v_z)}{g(v_z)v_z} = \text{constant}.$$

Letting this constant be denoted by $-b$ (the negative sign is included for convenience that will become apparent in a few lines), we find

$$\frac{\Phi'(|v|)}{\Phi|v|} = -b = \frac{g'(v_x)}{g(v_x)v_x}.$$

Using the explicit notation for derivative $g'(v_x) = \frac{dg(v_x)}{dv_x}$ and collecting all terms that contain v_x explicitly on the left-hand side, and all those that involve g on the right-hand side we get

$$-bv_x = \frac{dg(v_x)}{dv_x g(v_x)} \rightarrow -bv_x dv_x = \frac{dg(v_x)}{g(v_x)}.$$

We may integrate both side of this equation

$$-b \int v_x dv_x = \int \frac{dg(v_x)}{g(v_x)} \rightarrow -\frac{1}{2}bv_x^2 = \ln(g(v_x)) + c,$$

where c is the integration constant. Exponentiating both sides yields

$$\exp\left(-\frac{1}{2}bv_x^2\right) = g(v_x) \exp(c)$$

and renaming the constant factor $\exp(c) = A$ produces the final functional form for $g(v_x)$

$$g(v_x) = A \exp\left(-\frac{1}{2}bv_x^2\right).$$

We apply the same procedure to v_y and v_z and arrive at

$$P(v_x, v_y, v_z) = \Phi(|v|) = g(v_x)g(v_y)g(v_z) = A^3 \exp\left(-\frac{1}{2}b(v_x^2 + v_y^2 + v_z^2)\right) = A^3 \exp\left(-\frac{b}{2}|v|^2\right) \quad (6.3)$$

Finding the distribution parameters: 'A' and 'b'**A. Using the properties of probability distribution to infer the parameter 'A'**

Since A and $-b$ are physical quantities they must be determined. Recall that $g(v_x)$ is a probability function, therefore it must be normalized:

$\int_{-\infty}^{\infty} dv_x g(v_x) = A \int_{-\infty}^{\infty} dv_x \exp\left(-\frac{b}{2}v_x^2\right) = 1$; We may use this requirement to find A in terms of b .

The normalization problem can be solved by evaluating an important integral: the integral of a Gaussian function, a function of the form $\exp(-ax^2)$. You may choose to perform this integral using computer calculus programs. In these notes we also give a handy way of tackling Gaussian integrals analytically using pen and paper, a method that is attributed to a superstar physicist Richard Feynman.

First, let $I(a) = \int_{-\infty}^{\infty} \exp(-ax^2) dx \Rightarrow I^2(a) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp(-ax^2) \exp(-ay^2) dx dy = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dx dy \exp(-a(x^2 + y^2))$.

Then we perform a coordinate transform using the jacobian and convert x and y to polar coordinates by letting $x = r \cos(\theta)$ and $y = r \sin(\theta)$. The jacobian gives

$$dxdy = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} dr d\theta = \begin{vmatrix} \cos(\theta) & -r \sin(\theta) \\ \sin(\theta) & r \cos(\theta) \end{vmatrix} dr d\theta = (r \cos^2(\theta) + r \sin^2(\theta)) = r dr d\theta$$

and the vertical bars indicate taking a determinant and the leftmost matrix/determinant is the jacobian. Now,

$$I^2(a) = \int_0^{\infty} r dr \int_0^{2\pi} d\theta \exp(-ar^2) = 2\pi \int_0^{\infty} dr \cdot r \exp(-ar^2)$$

and letting $ar^2 = u \Rightarrow 2ar dr = du$ gives

$$I^2(a) = \frac{\pi}{a} \int_0^{\infty} \exp(-u) du = \left(\frac{\pi}{a}\right) \left(-\exp(-u)\big|_0^{\infty}\right) = \frac{\pi}{a}$$

taking square roots of both sides gives

$$\Rightarrow I(a) = \int_{-\infty}^{\infty} \exp(-ax^2) dx = \sqrt{\frac{\pi}{a}}$$

It follows that $\int_{-\infty}^{\infty} dv_x g(v_x) = A \int_{-\infty}^{\infty} dv_x \exp\left(-\frac{b}{2}v_x^2\right) = 1 \Rightarrow AI\left(\frac{b}{2}\right) = A\sqrt{\frac{2\pi}{b}} = 1 \Rightarrow$

$A = \sqrt{\frac{b}{2\pi}}$ so that $g(v_x) = \sqrt{\frac{b}{2\pi}} \exp\left(-\frac{b}{2}v_x^2\right)$.

B. Using the equipartition of energy theorem to infer the physical meaning of 'b'

Our final step is to find out what the parameter b stands for from the physical point of view. To do this we invoke the equipartition theorem which relates the statistical average of kinetic energy to temperature through a proportionality constant called the Boltzmann constant k_B (which is related to the gas constant R through $R = N_A * k_B$ where N_A is the number of particles in a mole of substances called the Avogadro number).

The *Equipartition of Energy Theorem* states that $\langle v_x^2 \rangle = \frac{RT}{M} = \frac{k_B T}{m}$ where $R = N_A k_B$ and $M = m N_A$ (N_A, k_B, m are Avogadro's number, Boltzmann constant and mass respectively).

$$\langle v_x^2 \rangle = \frac{k_B T}{m} = \int dv_x g(v_x) v_x^2 = \sqrt{\frac{b}{2\pi}} \int_{-\infty}^{\infty} dv_x v_x^2 \exp\left(-\frac{b}{2} v_x^2\right) = \frac{k_B T}{m}$$

Computing this integral using computer calculus is very easy, however, we would also like to show an alternative way of performing this integration using the *Feynman trick*:

$$I(a) = \int_{-\infty}^{\infty} \exp(-ax^2) dx = \sqrt{\frac{\pi}{a}} \Rightarrow \frac{dI(a)}{da} = \int_{-\infty}^{\infty} dx (-x^2) \exp(-ax^2).$$

By taking the derivative of $\sqrt{\frac{\pi}{a}}$ with respect to a we arrive at

$$-\frac{dI(a)}{da} = \frac{1}{2a} \sqrt{\frac{\pi}{a}}.$$

Returning to the Maxwell distribution we replace $a = \frac{b}{2}$, multiply by the normalization factor and obtain

$$\langle v_x^2 \rangle = \sqrt{\frac{b}{2\pi}} \int_{-\infty}^{\infty} dv_x (v_x)^2 \exp\left(-\frac{b}{2} v_x^2\right) = \frac{1}{b},$$

so that

$$\langle v_x^2 \rangle = \frac{1}{b} = \frac{k_B T}{m} \Rightarrow b = \frac{m}{k_B T} \Rightarrow b = \frac{m}{k_B T}.$$

So now using $b = \frac{m}{k_B T}$, we find the equation:

$$g(v_x) = \sqrt{\frac{m}{2\pi k_B T}} \exp\left(\frac{-m v_x^2}{2k_B T}\right)$$

Performing the same steps for $g(v_y), g(v_z)$ and recalling Assumption 1,

$$P(v_x, v_y, v_z) = g(v_x)g(v_y)g(v_z) = \left(\frac{m}{2\pi k_B T}\right)^{\frac{3}{2}} \exp\left(\frac{-m(v_x^2 + v_y^2 + v_z^2)}{2k_B T}\right) = \left(\frac{m}{2\pi k_B T}\right)^{\frac{3}{2}} \exp\left(\frac{-m|v|^2}{2k_B T}\right) \quad (6.4)$$

The absolute value for $|v|$ may of course be dropped since the speed is squared in the final expression but the fact that it is squared reflects that Assumption 2 is respected by this expression: v^2 is perfectly isotropic. This is the well-known Maxwell-Boltzmann distribution.

Chapter 7

Ensemble Theory

The methods of statistical mechanics are concerned with predicting behaviors of matter starting from 'microscopic' or 'molecular' details. Macroscopic systems typically contain on the order of Avogadro number of particles and direct atomistic simulations are not feasible. Instead, in order to predict properties of interest we resort to statistical approaches, we model the probability distribution of system's variables and compute the statistical expectation value of the relevant observable. For instance, the quantity that is referred to as 'internal energy' in thermodynamics is simply the expectation value of the energy of the system with respect to the equilibrium distribution $P(x, p)$:

$$\langle E \rangle = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} E(x, p) P(x, p) dx dp$$

where x is the vector of positions and p is vector the momenta (momentum is defined as the product of the particle's mass m and velocity v , $p = mv$). This places the entire theoretical burden on finding the probability distribution that described the system we are interested to analyze. The objective of **ensemble theory** is to do precisely that: to identify the general features of probability distributions for three classes of systems: P_{NVE} , P_{NVT} , and $P_{VT\mu}$. The classes of systems that we will consider are:

1. A closed system in which the volume (V), energy(E), matter(N) are conserved: NVE or Microcanonical ensemble
2. A system in contact with a heat-reservoir at temperature T in which the volume and matter are conserved: NVT or Canonical ensemble
3. A system in contact with a heat-reservoir that exchanges energy and matter with surroundings: Grand canonical ensemble.

One key reason why it is important to understand ensemble theory very well is because it provides the foundation for everything we do in statistical mechanics. Ensemble theory is the gateway to computational molecular science.

What is an ensemble? *Ensemble*: Consider N particles in a volume V with energy E. Particles are labelled using the index i and $i = 0, 1, 2, \dots, N$ where the i -th particle has a position

coordinate denoted q_i of three components (x, y, z) and a momentum coordinate denoted p_i also of three components (p_x, p_y, p_z) . Taken together for the N particles, they define a $6N$ dimensional space called *phase space* which is denoted with the Greek letter Γ . The combined set of configurations of all these particles in phase-space is called an ensemble. We may imagine some molecular system in one particular configuration as being a member of the ensemble. Then we replicated it Ω times such that each replica contains a distinct configuration of positions and momenta of the particles. One way to think of this is to consider snapshots of a molecular system as it is moving around: each snapshot represents a point in the phase-space Γ with specific values of positions of all atoms and also their momenta.

Now, since what determines the dynamics in classical Newtonian mechanics is the energy function: the sum of kinetic and the potential energies for all particles, then energy will determine to a large degree which configurations or points in phase space are accessible to our molecular system. If the energy is fixed, and the number of particles does not change, and the physical space in which our system 'lives' (the volume) is fixed as well: fixed energy condition constrains our ensemble to a very well defined set of points in the phase-space - the constant energy 'hyper-shell'.

An important axiom that we will use states that all points in this constant-energy space are equally likely to be visited. This axiom is called the **equal a-priori probability**. Another axiom that we will use later on is called the **ergodic hypothesis**: An ergodic system, if we let it evolve in time long enough, will sample all points in phase space that it is allowed to sample given the constraints of fixed volume, energy, and number of particles. This axiom is needed for computing expectation values using dynamical trajectories, it tells us that if we have a long enough trajectory the expectation values computed using the distribution sampled by this trajectory are equivalent to the expectation values computed using a phase space ensemble (as defined above).

We are now in a position to tackle the first class of systems: the closed systems for which N, V, E are fixed, i.e., we are dealing with a **microcanonical ensemble**. In this case, the phase-space trajectories visit points in a constant energy *hypersphere* and equal probability is assigned to every configuration of the system whose phase-space coordinates fall in $E(q, p) + \Delta E$ (with small ΔE). We will use the symbol Ω for the count of all possible configurations that fall into the fixed energy 'hypershell'. The 'equal a-priori probability' axiom gives us the probability distribution for states in microcanonical ensemble: it is the uniform distribution with $P = \frac{1}{\Omega}$ for all states.

This may seem to be an insignificant development, after all we have simply recovered the postulated axiom. However, we will use the properties of the microcanonical ensemble to find the properties of the canonical and the grand canonical ensembles. To do that we will need to know how to calculate (with the goal of eventually maximizing) the entropy in the microcanonical ensemble. We will obtain this useful result now: we'll show that it is given by the familiar Boltzmann Entropy starting from the more general expression called the Gibbs Entropy: $S = -k_B \sum_{i \in \Omega} p_i \log(p_i)$. Since all states are equally probable the distribution is uniform, i.e., $p_i = \frac{1}{\Omega}$ (assuming equal probabilities; in the absence of degeneracies i.e. all states are unique),

$$S = -k_B \sum \frac{1}{\Omega} \log\left(\frac{1}{\Omega}\right) = k_B \log(\Omega) \quad (7.1)$$

Next we will present and apply the maximum entropy approach to the canonical ensemble and derive P_{NVT} .

Canonical Ensemble:

Next we will present and apply the maximum entropy approach to the canonical ensemble and thereby derive the general form for P_{NVT} . Define a generic system with a fixed number of particles N confined in a fixed volume V , and construct an ensemble by putting copies of this system in contact with each other. We will keep the number of particles and the volumes fixed, just like in the microcanonical ensemble, but we will allow the energy to be exchanged between the different copies of our system. The total energy of this ensemble will remain conserved, so overall the whole system plays by the rules of the micro-canonical ensemble. We will shortly see that maximizing the entropy of this system will give rise to the concept of temperature.

-ensemble: create n copies of system:

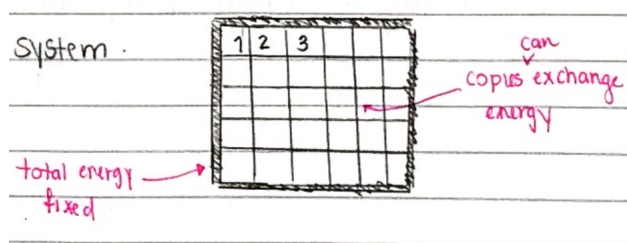


Figure 7.1: ' n ' copies of the system that can exchange energy, however, the total energy is fixed.

-total energy: E

-possible energy states: $E_1, E_2, \dots$

Then have n_1 copies of E_1 , n_2 copies of E_2 , etc... until have n such copies so that

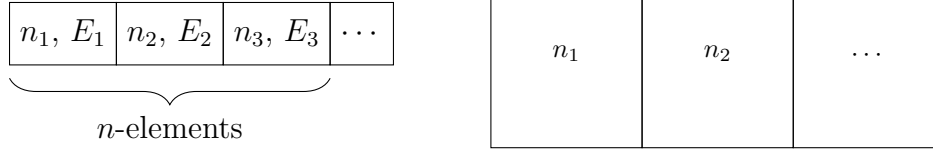
$$\sum_i n_i = n$$

and

$$\sum_i n_i E_i = E$$

What we are missing is a quantitative form for the number of states Ω . In order to find a convenient expression for Ω

, let's sort the ensemble into bins according to the 'microstate's' energy: -set of copies n_1 has energy E_1 , n_2 has E_2 , etc.



Since we want to sort the n objects (recall $\sum n_i = n$) into (distinguishable) boxes such that n_1 go in box 1, n_2 go in box 2, etc.

The number of arrangements is

$$\Omega = \frac{n!}{n_1! n_2! \dots} = \frac{n!}{\prod_i n_i!}$$

Now we are in a position to act on our plan and to maximize $\log(\Omega)$ subject to

$$\sum n_i = n$$

and

$$\sum n_i E_i = E$$

. First, recall from logarithm rules that

$$\log \frac{n!}{\prod_i n_i!} = \log(n!) - \log\left(\prod_i n_i!\right)$$

, and recall Sterling's approximation,

$$\log(n!) \approx n \log(n) - n$$

(as n gets larger, the approximation becomes better, see assignment 2 question 1A). Then

$$\Rightarrow \log(\Omega) = n \log(n) - n - \sum n_i \log(n_i) + \sum n_i = n \log(n) - \sum n_i \log(n_i)$$

where the first condition $\sum n_i = n$ was used and Sterling's approximation was used on both factorial terms, $\log(n!)$ and $-\log\left(\prod_i n_i!\right)$.

Lagrange Multipliers α and β will now be used. Let

$$Y = n \log(n) - \sum n_i \log(n_i) - \alpha \sum n_i - \beta \sum n_i E_i$$

We then find the extrema of this function w.r.t n_i . At the maximum the differential vanishes, so we set

$$dY = -dn_i - \log(n_i)dn_i - \alpha dn_i - \beta E_i dn_i = (-1 - \log(n_i) - \alpha - \beta E_i)dn_i = 0$$

$$\Rightarrow \log(n_i) = -1 - \alpha - \beta E_i \Rightarrow n_i = \exp(-1 - \alpha) \exp(-\beta E_i)$$

Letting $A = \exp(-1 - \alpha)$,

$$\Rightarrow P_i = \frac{n_i}{\sum n_i} = \frac{e^{-\beta E_i}}{\sum_i e^{-\beta E_i}}$$

The quantity $\sum_i e^{-\beta E_i}$ is extremely important and is called the *partition function*, commonly denoted

$$\sum_i e^{-\beta E_i} = Z \tag{7.2}$$

Now recall for probability P , $\sum_i P_i = 1$ so that $\Rightarrow \sum_i dP_i = 0$. Then

$$\bar{E} = \sum_i P_i E_i = \sum_i \frac{e^{-\beta E_i}}{\sum_j e^{-\beta E_j}} E_i = \frac{1}{Z} \sum_i e^{-\beta E_i} E_i$$

since i and j are just indices and can be re-labelled.

Chapter 8

Connection to Thermodynamics

We have $dU = TdS - pdV$ (note that pressure here is denoted as lower-case p , since P refers to probability here). Then $U = \bar{E} = \frac{1}{Z} \sum_i e^{-\beta E_i} E_i$, therefore

$$dU = d\bar{E} = d \sum P_j E_j = \sum (dP_j E_j + P_j dE_j)$$

First, note that $P_j = \frac{1}{Z} e^{\beta E_j} \rightarrow E_j = \frac{-1}{\beta} (\log(P_j) + \log(Z))$ which is obtained from taking logarithms and then rearranging. Next we deal with the sum over the two differentials. Let's say $dE_{1'} = \sum dP_j E_j$ and $dE_{2'} = \sum P_j dE_j$. Then

$$dE_{1'} = \left(\frac{-1}{\beta} \sum_j \log(P_j) dP_j - \frac{1}{\beta} \log(Z) \sum_j dP_j \right) = \frac{-1}{\beta} \sum_j \log(P_j) dP_j$$

since the second term is 0 (since $\sum_j dP_j = 0$). Now we use $-\frac{1}{\beta} \sum_j \log(P_j) dP_j = -\frac{1}{\beta} d \left(\sum_j P_j \log(P_j) \right)$ which is justified since

$$\begin{aligned} -\frac{1}{\beta} d \left(\sum_j \log(P_j) P_j \right) &= -\frac{1}{\beta} \sum_j \left(\log(P_j) dP_j + d(\log(P_j)) P_j \right) \\ \Rightarrow -\frac{1}{\beta} \sum_j \left(\log(P_j) dP_j + \left(\frac{1}{P_j} \right) P_j dP_j \right) &= -\frac{1}{\beta} \sum_j \log(P_j) dP_j \end{aligned}$$

where we brought the differential d inside and used the product rule (again, $\sum_j dP_j = 0$).

Therefore we note that

$$dE_{1'} = -\frac{1}{\beta} d \left(\sum_j P_j \log(P_j) \right)$$

Next for $E_{2'}$:

$$E_{2'} = \sum P_j dE_j = \sum P_j \frac{\partial E_j}{\partial V} dV$$

Where we note that $dE_j(V) = \frac{\partial E_j}{\partial V_j} dV$. Finally, by interpreting

$$\sum_j P_j \frac{\partial E_j}{\partial V} = -p$$

as the thermodynamic pressure p , we arrive at the conclusion

$$dU = d\bar{E} = dE_{1'} + dE_{2'} = -\frac{1}{\beta} d\left(\sum_j P_j \log(P_j)\right) + \sum_j P_j \frac{\partial E_j}{\partial V_j} dV = TdS - pdV \quad (8.1)$$

Recall that Gibbs Entropy is given by the formula is $S = -k_B \sum_j P_j \log(P_j)$ (where k_B is the Boltzmann constant and P_j is the probability in the j -th state. We are now in a position to identify the Lagrange multiplier β by using the identity $\frac{1}{\beta k_B} = T$ which gives us $\beta = \frac{1}{k_B T}$ (where T is temperature).

Helmholtz Free Energy:

$$\begin{aligned} A = U - TS &= \sum_i P_i E_i - \frac{1}{k_B \beta} k_B \sum_j P_j \log(P_j) \\ \Rightarrow \sum_j \frac{e^{-\beta E_j}}{Z} E_j + \frac{1}{\beta} \sum_j \frac{e^{-\beta E_j}}{Z} [-\beta E_j - \log(Z)] &= \sum_j \frac{e^{-\beta E_j}}{Z} E_j - \frac{1}{\beta} \sum_j \frac{e^{-\beta E_j}}{Z} (\beta E_j) - \frac{1}{\beta} \sum_j \frac{e^{-\beta E_j}}{Z} \log(Z) \end{aligned}$$

where we have used $\log(P_j) = -\beta E_j - \log(Z)$, canonical distribution P , $\beta = \frac{1}{k_B T}$, etc. The first two terms in the last step cancel, so

$$\Rightarrow -\frac{1}{\beta} \sum_j \frac{e^{-\beta E_j}}{Z} \log(Z) = -\frac{1}{\beta} \frac{\log(Z)}{Z} \left(\sum_j e^{-\beta E_j} \right) = -\frac{1}{\beta} \frac{\log(Z)}{Z} (Z) = -\frac{1}{\beta} \log(Z)$$

where we observe that $\log(Z)$ and Z are simply numbers and can be taken out of the sum, leaving another Z that cancels with the denominator one. Therefore

$$A = -\frac{1}{\beta} \log(Z) \quad (8.2)$$

We can now make the important observation from the canonical ensemble that

$$\bar{E} = -\frac{1}{Z} \left(\frac{\partial Z}{\partial \beta} \right)_{N,V} = -\left(\frac{\partial \log(Z)}{\partial \beta} \right)_{N,V} = \left(\frac{\partial \beta A}{\partial \beta} \right)_{N,V} \quad (8.3)$$

Similar approaches may be used to find other energies.

Heat Capacity (constant Volume):

$$C_V = \left(\frac{\partial U}{\partial T} \right)_{N,V} = \frac{\partial}{\partial T} \frac{\partial \beta A}{\partial \beta} = -k_B \beta^2 \frac{\partial^2 \beta A}{\partial \beta^2}$$

where we have used (8.3) and the fact that

$$\frac{\partial}{\partial T} = \frac{\partial \beta}{\partial T} \frac{\partial}{\partial \beta} = -\frac{1}{k_B T^2} \frac{\partial}{\partial \beta} = -\frac{k_B}{k_B^2 T^2} \frac{\partial}{\partial \beta} = -k_B \beta^2 \frac{\partial}{\partial \beta}$$

Using (8.3) again we next find

$$C_V = k_B \beta^2 \frac{\partial^2 \log(Z)}{\partial \beta^2} = k_B \beta^2 \left[-\frac{1}{Z^2} \frac{\partial Z}{\partial \beta} \frac{\partial Z}{\partial \beta} + \frac{1}{Z} \frac{\partial^2 Z}{\partial \beta^2} \right] = k_B \beta^2 \left[\frac{1}{Z} \frac{\partial^2 Z}{\partial \beta^2} - \left(\frac{1}{Z} \frac{\partial Z}{\partial \beta} \right)^2 \right] = k_B \beta^2 \left[\frac{1}{Z} \frac{\partial^2 Z}{\partial \beta^2} - (\bar{E})^2 \right]$$

Remember, $\frac{\partial(\log(f(x)))}{\partial x} = \frac{1}{f(x)} \frac{\partial f(x)}{\partial x}$. We now evaluate the first term in the brackets:

$$\frac{1}{Z} \frac{\partial^2 Z}{\partial \beta^2} = \frac{1}{Z} \frac{\partial^2}{\partial \beta^2} \left[\sum_j e^{-\beta E_j} \right] = \frac{1}{Z} \frac{\partial}{\partial \beta} \left[\sum_j (-E_j) e^{-\beta E_j} \right] = \frac{1}{Z} \left[\sum_j (-E_j)^2 e^{-\beta E_j} \right] = \sum_j P_j (E_j)^2 = \overline{(E^2)}$$

this means that

$$C_V = k_B \beta^2 [\overline{(E^2)} - (\bar{E})^2] = k_B \beta^2 [\langle E^2 \rangle - \langle E \rangle^2] = k_B \beta^2 \langle \Delta \bar{E} \rangle^2 \quad (8.4)$$

which shows that the heat capacity captures the in fluctuations of internal energy of a system at a fixed temperature T.

Chapter 9

The Maximum Entropy Principle for data-driven modelling

In 1957, famed physicist Edwin Jaynes pointed out in an important paper the relation between statistical mechanics and information theory, today seen in "big data".

Recall that in order to derive the canonical distribution,

$$P(E_i) = \frac{e^{-\beta E_i}}{\sum_i e^{-\beta E_i}}$$

we *maximized* the **entropy under constraints of our ensemble**, driven by the physical notion that at equilibrium entropy is maximal.

From the information theory point of view, this is equivalent to developing the *least biased model*. This is the same as declaring absolute ignorance about aspects of the system for which we do not have experimental data.

For example, let's look at an example with chromosomes:

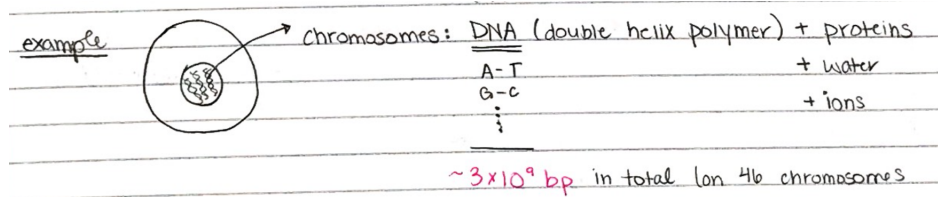


Figure 9.1: The example we will use for DNA in chromosomes.

-distance between base-pairs is about $0.5nm$; one chromosome is approximately $5cm$ if stretched out

-Molecular Dynamics is not likely to be used here because the system is too large

-instead we will take experimental organization of chromatin in the nuclei and use the **Maximum Entropy Principle** to build a model; here we will use a generic homopolymer and constrain it using experimental information

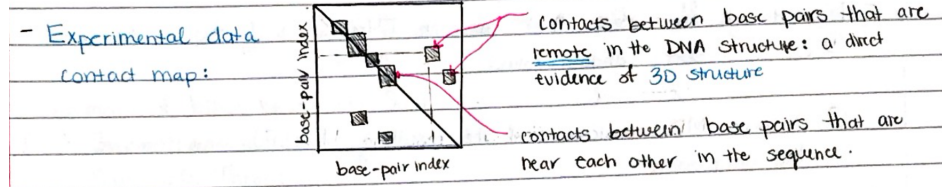


Figure 9.2: Contact map for our example.

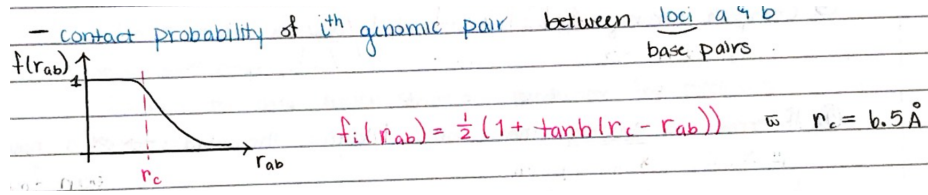


Figure 9.3: Radial distribution with contact probability for our example.

$$f_i(r_{ab}) = \frac{1}{2}(1 + \tanh(r_c - r_{ab}))$$

with $r_c = 6.5 \text{ \AA}$

Maximum Entropy distribution of a generic homopolymer under the constraint of contact probabilities $f_i(r_{ab})$:

$$P(U_{ME}(r)) = \frac{e^{-\beta U_{ME}(\vec{r})}}{Z}$$

where $U_{ME} = U(r) + \sum \alpha_i f_i(r_{ab})$ and α_i are Lagrange multipliers that need to be found.

How do we find these α_i parameters? We first need to define a *cost function*. This cost function reflects the "distance" between the distributions.

$$\Gamma(\alpha) = \log \left(\frac{Z_\alpha}{Z_0 e^{-\beta \sum \alpha_i f_{i, \text{experiment-data}}}} \right)$$

where Z_0 is the partition function of the homopolymer and Z_α is the partition function of $U_{ME}(r)$,

$$e^{\Gamma(\alpha)} = \frac{Z_\alpha}{Z_0 e^{-\beta \sum \alpha_i f_{i, \text{experiment-data}}}} \Rightarrow Z_0 e^{-\beta \sum \alpha_i f_{i, \text{experiment-data}}} = e^{-\Gamma(\alpha)} Z_\alpha$$

Now note we want $\Gamma(\alpha) \rightarrow 0$. So the question now is how do we find α 's that minimize $\Gamma(\alpha)$?

1. Set $\vec{\alpha}_0$ some random values
 2. Perform MD simulations of the resulting $U_{ME}(\vec{r})$ and compute $\langle f_i \rangle$ from the simulation
 3. Compute error: $\frac{\sum_i |\langle f_i \rangle - f_{i,exp}|}{\sum_i f_{i,exp}}$ where *exp* stands for experimental data
 4. If error is smaller than some tolerance threshold then α has been found.
- If error is larger than the threshold then updates the α 's:

$$\alpha_{next-step} = \alpha_{pre-step} + \frac{1}{B\beta}(\langle f_i \rangle - f_{i,exp})$$

and then go back to step 2.

Chapter 10

Spin systems

10.1 Hopfield Network and the Ising model

The Hopfield network is a system at the boundary of statistical mechanics and machine learning. Systems consist of components that take on one or two values such as $\{0, 1\}$ or $\{-1, 1\}$ - these may be interpreted as pointing up/down and for this reason on Physics and in Chemistry they are called 'spins'.

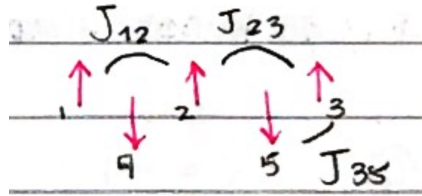


Figure 10.1: An illustration of a system of 5 interacting spins

We energy function for this collection of spins may be written as follows (we will encounter it again when we discuss the Ising Model):

$$E = - \sum h_i s_i - \sum_{\langle ij \rangle} J_{ij} s_i s_j \quad (10.1)$$

for $s_i \in \{0, 1\}$, s_i is the state of the i -th spin and $\langle ij \rangle$ denotes all pairs ij .

The Hopfield network dynamics consists of sequential flipping of spins with flips being accepted if the total energy of the spin-system is lower after the flip than it was before. Performing this flips will eventually lead us to a local (or global) minimum of E .

$$\Delta E_i = E(s_i = 0) - E(s_i = 1) = h_i + \sum_{\langle ij \rangle} s_i J_{ij}$$

where the h_i 's often can be set to zero for simplicity.

Example. Find the minimum energy for:

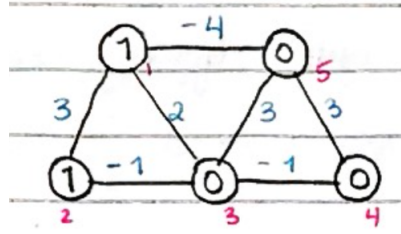


Figure 10.2: How to draw/write out a Hopfield network diagram.

$$-E = (1 \cdot 1 \cdot 3) + (1 \cdot 0 \cdot 2) + (1 \cdot 0 \cdot -1) + (1 \cdot 0 \cdot -4) + (0 \cdot 0 \cdot 3) + (0 \cdot 0 \cdot -1) + (0 \cdot 0 \cdot 3) = 3$$

so that $E = -3$

Step 1: Check whether flipping state number 5 lowers the total energy.

$$E = -(1 \cdot 1 \cdot 3 + 1 \cdot 1 \cdot 4) = 1 \Rightarrow$$

do not flip.

Step 2: Check whether flipping state number 2 lowers the total energy.

$$E = 0 \Rightarrow$$

do not flip.

Step 3: Check whether flipping state number 3 lowers the total energy.

$$E = -(1 \cdot 1 \cdot 3 + 1 \cdot 1 \cdot 2 + 1 \cdot 1 \cdot -1) = -4 \Rightarrow$$

accept the flip!

Continue on in this way and when we find that we will not accept any of the other steps then we have found a *local minimum* at $E = -4$.

There is also another minimum found at $E = -5$

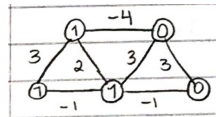


Figure 10.3: A local minimum for $E = -4$

For data-driven modelling the goal is to find the values of J 's that will encode a desired energy landscape (i.e the locations of the minima).

Chapter 11

The Ising Model

Non-interacting Spins:

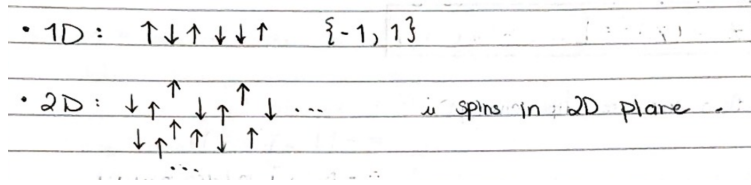


Figure 11.1: Ising model in 1-D and 2-D for non-interacting spins.

Recall from (10.1) that $E = \sum h_i s_i + \sum -\mu B \cdot s_i$ with $s_i \in \{-1, 1\}$.

By convention, we consider a uni-axial magnetic field $B = B_z$.

The energy function contains only partial information about the system. We need the **free energy** instead.

Consider the **canonical distribution** (b/c the number of spins is assumed to be fixed). Then

$$Z = \sum_i e^{-\beta E_i} = \sum_{s_1=\pm 1} \sum_{s_2=\pm 1} \cdots \sum_{s_N=\pm 1} e^{-\beta(\sum h_i s_i - \sum \mu B \cdot s_i)}$$

where i is the number of microstates. For simplicity, assume $\vec{h} = 0$, then

$$Z = \sum_{s_1=\pm 1} \sum_{s_2=\pm 1} \cdots \sum_{s_N=\pm 1} e^{\beta(\sum \mu B \cdot s_i)} \Rightarrow$$

By breaking this sum into its N components we find that

$$= \sum_{s_1=\pm 1} e^{\beta(\sum \mu B \cdot s_1)} \sum_{s_2=\pm 1} e^{\beta(\sum \mu B \cdot s_2)} \cdots \sum_{s_N=\pm 1} e^{\beta(\sum \mu B \cdot s_N)} = \left[\sum_{s=\pm 1} e^{\beta \mu B \cdot s} \right]^N = (Z_1)^N$$

where N in this case is the number of spins and the 1 in Z_1 just stands for the number of possible spin "directions" for a **single/one** spin (remember in our case we are saying one direction is +1 so the opposite direction is labelled -1). Now for our current example we have two possible choices for a single spin, -1 and 1 (remember the labelling is arbitrary,

we could easily choose the spins to be something else like 0 and 1). So for a single spin with two possible choices of spin "direction", we have

$$Z_1 = e^{\beta\mu B(-1)} + e^{\beta\mu B(+1)} = 2 \cosh(\beta\mu B)$$

(recall the definition of hyperbolic trig functions, eg. $\cosh(x) = \frac{1}{2}(e^x + e^{-x})$). Thus for N number of spins,

$$Z_N = [2 \cosh(\beta\mu B)]^N$$

which is an explicit calculation of the partition function here!

We can calculate the **free energy** with

$$F = -k_B T \ln(Z_N) = -N k_B T \ln(Z_1) = -N k_B T \ln\left(2 \cosh\left(\frac{\mu B}{k_B T}\right)\right)$$

and the **average energy** with

$$\bar{E} = -\frac{\partial \ln(Z_N)}{\partial \beta} = \frac{\partial F}{\partial \beta} = \dots = -N \mu B \tanh(\beta\mu B)$$

where the "... " means we have used the chain rule. Graphically,

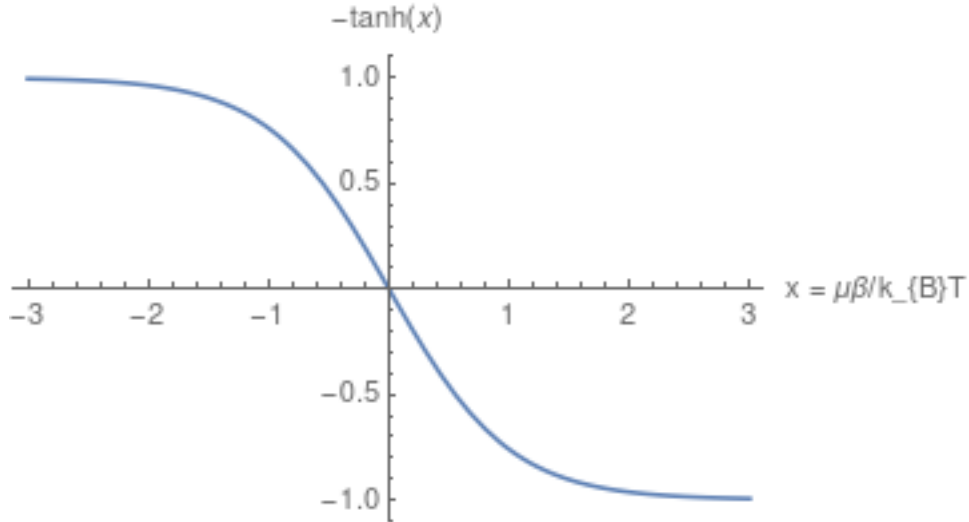


Figure 11.2: Graph of $-\tanh\left(\frac{\mu B}{k_B T}\right)$

Now, is it that entropy or energy profile that dominates the system? At high temperatures, entropy dominates because the spins become randomly oriented while at low temperatures there's some order of the spins due to the presence of the magnetic field, etc.

11.1 Observables

It is time to note some interesting characteristics that can be measured and that are useful in the study of statistical mechanics. Two observables we will define are **magnetization**

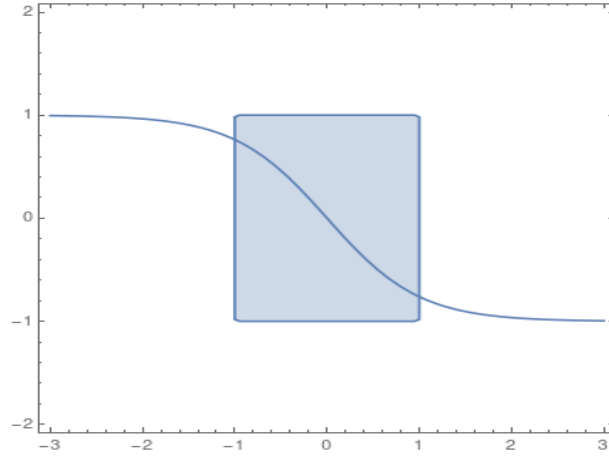


Figure 11.3: Graph of $-\tanh\left(\frac{\mu B}{k_B T}\right)$, where the highlighted region (shaded blue) is the high temperature region. At high temperatures, $\mu B \ll k_B T$ (or $\frac{\mu B}{k_B T} \ll 1$) while at low temperatures, $\mu B \gg k_B T$ (or $\frac{\mu B}{k_B T} \gg 1$).

and **susceptibility**.

Magnetization:

$$\bar{M} = \mu \sum_{i=1}^N \bar{s}_i = k_B T \frac{\partial \ln Z}{\partial B} = -\frac{\partial F}{\partial B} = N\mu \tanh(\beta\mu B) \quad (11.1)$$

Isothermal Susceptibility:

$$\chi = \left(\frac{\partial \bar{M}}{\partial B} \right)_T \quad (11.2)$$

where the partial derivative can be interpreted as a response to a small magnetic field.

By Taylor series expansion about $B = 0$,

$$\bar{M} = \frac{1}{0!} \bar{M}(B=0) + \frac{1}{1!} \frac{\partial \bar{M}}{\partial B} dB + \frac{1}{2!} \frac{\partial^2 \bar{M}}{\partial B^2} dB^2 + \dots$$

and for vanishingly small dB of terms of order 2 and higher, the series may be truncated to just the first two terms. Therefore for small B ,

$$M \approx \text{constant} + \chi dB$$

where we have used (11.2) for the second term. By performing the series expansion we find that

$$\chi = N\beta\mu^2$$

which may be re-written in a form referred to as the Curie form for isothermal magnetic susceptibility

$$\chi = \frac{C}{T}$$

where $C = \frac{N\mu^2}{k_B}$ which is an important result in high temperature magnetism ($k_B T \gg \mu B$).

Chapter 12

Ising Model with Interactions

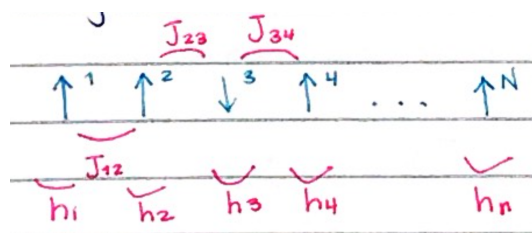


Figure 12.1: A schematic representation of interacting spins.

Here we will 'scratch the surface' of the field of interacting spin models by analyzing a simple 1 dimensional (1D) chain of spins that interact only with nearest neighbors via a fixed interaction parameter J . The energy function for this spin chain is simply given by

$$E = -J \sum s_i s_j$$

where $s_i = \pm i$. We would like to explore some of the physical properties of this spin system, such as the behavior of magnetization, heat capacity, magnetic susceptibility, etc... To do this we need to derive the Helmholtz free energy (for which symbols F or A are used), $A = F = -k_B T \ln(Z)$, and to that end we need to compute the partition function Z .

Computing the partition function for a thermodynamic system of interacting spins in general is extremely challenging. Here, because of our simplifying choices for interactions and in the absence of dependence on a magnetic field we will be able to find an easy solution by constructing a recursive expression and unfolding it down to the trivial case of the partition function for only a single particle Z_1 . Other methods are required for more general cases but our goal here is to provide only a glimpse into the field and we will not attempt to make the presentation of the topic complete.

Let's start by fixing the number of spins N and writing out explicitly the partition functions for small numbers of spins: $N = 2$ and $N = 3$ using the canonical (NVT) ensemble. We will try to notice a pattern and use it to our advantage. Our plan is two-fold

Step 1. Compute Z for *finite* N .

Step 2. Extrapolate Z_N to $N \rightarrow \infty$.

* For $N = 2$ there are 4 microstates:

$$\begin{array}{cccc} \uparrow \downarrow & \uparrow \uparrow & \downarrow \uparrow & \downarrow \downarrow \\ \text{config. 1} & \text{config. 2} & \text{config. 3} & \text{config. 4} \end{array}$$

So that the energy in config. 1 is $E_{\text{config.1}} = +J$, $E_{\text{config.2}} = -J$, $E_{\text{config.3}} = +J$ and $E_{\text{config.4}} = -J$ which is found by using (??). For example, using *config.1*, *config.1* = $-J(+1)(-1) = +J$ (remember, spin up = +1 while spin down = -1) so that

$$Z = e^{-\beta J} + e^{-\beta J} + e^{\beta J} + e^{\beta J}$$

and so

$$Z_2 = 2e^{\beta J} + 2e^{-\beta J} = 4 \cosh(\beta J)$$

* For $N = 3$ there are 8 micro-states:

$$\begin{array}{ccc} \uparrow \downarrow \uparrow & \uparrow \uparrow \uparrow & \\ \text{config. 1} & \text{config. 2} & \text{'...'} \end{array}$$

$$Z_3 = 2e^{2\beta J} + 4 + 2e^{-2\beta J} = 4 \cosh(\beta J) = 2(e^{\beta J} + e^{-\beta J})^2 = (e^{\beta J} + e^{-\beta J})Z_2$$

This suggests a recursion relation:

$$Z_N = 2 \cosh \beta J Z_{N-1} = (2 \cosh \beta J)^2 Z_{N-2} = (2 \cosh \beta J)^3 Z_{N-3} = (2 \cosh \beta J)^{N-1} Z_1.$$

Z_1 is of course trivial, the energy is zero for a system with only one spin (since only interactions with other spins modify the energy function in our case) and summing over the two possible configurations of a two-state spin $Z_1 = (e^0 + e^0) = 2$. We conclude that the partition function for N spins is give by

$$Z_N = 2(2 \cosh \beta J)^{N-1}.$$

We are now in a position to obtain the free energy for N spins using the definition

$$F_N - k_B T \ln Z_N = -k_B T [\ln(2) + (N-1) \ln \left(2 \cosh \left(\frac{J}{k_B T} \right) \right)].$$

We would like to learn about a thermodynamic system, which means that we need to send the number of spins to infinity. When we take the limit of $N \rightarrow \infty$ the terms in F_N that do not depend on N become negligible, the difference between N and $N-1$ vanishes and the expression simplifies to

$$F_\infty = F = -N k_B T \ln \left(2 \cosh \left(\frac{J}{k_B T} \right) \right) = -\frac{N}{\beta} \ln(2 \cosh(\beta J)).$$

The last thing I would like to discuss is the intuition behind phase transitions and why 1D Ising model fails to predict a magnetic phase transition.

Consider a chain of N ferromagnetic spins ($J > 0$, the energy is lowered when neighboring spins are pointing in the same direction) all pointing upwards. The energy of this chain at

$T = 0K$ is $E = -(N - 1)J$ and the entropy is $S = 0$ (in the absence of a magnetic field to break isotropy of space the direction up/down is not meaningful, the only thing that matters is whether the spins are pointing in the same direction or in opposite directions).

$\uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \dots \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow$

Illustration: 1D chain of ferromagnetic spins in ground state at $T = 0K$

Once temperature departs from the absolute zero, excitations from the minimum energy ground become possible. Consider an excitation of the particular type: a domain flip. A domain flip is when an entire block of spins flips at once. An example of a domain flip is shown below.

$\downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \dots \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow$

An example of a spin-chain configuration following a single domain flip

The energy cost of this excitation is not very high, it can be shown to be only $2J$. The number of possible domain boundaries in a chain of N spins is $N - 1$, so the entropy change in this process is given by $k_B \ln(N - 1)$, and the free energy $dF = dU - TdS = 2J - k_B T \ln(N - 1)$. Sending N to the thermodynamics limit ($N \rightarrow \infty$) we see that even for very small temperatures the entropy term will turn the balance of free energy for domain flipping into the negative very easily and so a meaningful temperatures for which the phase transition happens from the ordered to the disordered magnetic states cannot be found for 1D Ising chain in the absence of a magnetic field. It may be found for a 1D Ising chain in the presence of a magnetic field and for high-dimensional Ising models, but we leave the exploration of magnetic phase transitions to more advanced classes.

Chapter 13

Theory of polymers

Polymers are long chains of covalently bonded monomers. Polymers are commonly used to make materials with unique properties. Much of what we see around ourselves from furniture to electronic devices is made by chemically designing polymers to exhibit the required properties: hardness, density, melting temperature, heat capacity, electrical conductivity, heat conductivity etc... Much of what we see around ourselves is made out of polymers with static properties, polymers that do not respond in a meaningful way to changes in surroundings. Beyond these common materials, a fascinating new class of materials based on polymers has recently emerged: these are self-healing materials, for example conducting polymer wires that we can cut with a knife only to observe the electric current resume after a short while. Biological polymers are commonly used at the boundary between materials and biology, for example, single stranded DNA polymers are used as highly sensitive chemical and biochemical sensors, peptides and proteins are used to guide nano-particle synthesis and to form bio-mimetic materials with self-healing and adhesive properties. Finally, biological polymers play fundamentally important roles in biology: from proteins to RNA and DNA sequences to complicated super-aggregates of these polymers forming the genetic material called chromatin, polymers are ubiquitous in biology and significant efforts are directed to understand the interplay between their physical properties as polymers and their biochemical behaviors in living cells.

In this chapter we will explore basic methods for theoretically describing the conformation of polymer chains. Here, we will focus our attention on ideal polymers. Ideal polymers are toy representations of polymers in the same way ideal gas is a toy representation of real gases. Ideal polymers capture some of the basic characteristics of real polymer but they are not burdened by the specific properties of any individual polymer - this abstraction makes the math easy to handle and this makes the conclusions general, at some level we expect most polymers to behave like toy representations. The key assumption of course is that intramolecular steric interactions are insignificant: the polymer is allowed to intersect itself without paying energetic penalty (much like ideal gas particles are allowed to occupy the same space without energy rising). Just like with ideal gas, this assumption is not physical, but we will find out that it is reasonable in many cases and we can learn a lot from this simplified model of a polymer.

13.1 Toy models of polymers

In our description of polymers we strip them of chemical identities and think for monomers as beads linked together in a chain.

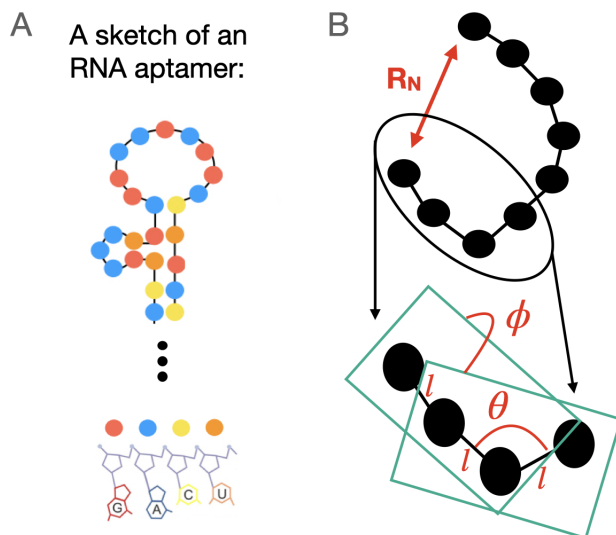


Figure 13.1: Illustration of toy representations of polymers. (A) A schematic representation of a folded RNA aptamer with color-coded bases; Legend is shown on the figure. (B) A toy representation of a generic homopolymer with monomers represented by black beads. The end-to-end distance R_N , the distance between monomers l , the angle θ between adjacent intermonomer 'bonds', and the dihedral angle ϕ are visually defined on the figure.

Polymers typically coil up and fold in solution driven by entropic and energetic forces. For this reason, large distance between monomers in sequence may not correspond to a large distance in real space: distant residues in sequence may form a contact in the 3D folded structure of the polymer. By way of an illustrative example consider the fact that a single human chromosome when stretched out is around 5cm long yet it fits into microscopic nucleus inside the human cells - the overall size of a polymer may be reduced by several orders of magnitude in the folded and packed configuration. For this reason one of key observables we may wish to explore is the end-to-end distance R_N and the radius of gyration for a polymer. We may ask ourselves, for example, how would the folded size depend on the number of monomers? Next we will develop two simple models that will help us answer this question in minimalistic fashion: we will try to include as few details about the polymers and learn only very generic features and behaviors.

13.1.1 End-to-end distance of polymers

Our first question is: What happens to the size of the coiled up polymer as its length increases?

To answer this question we need to define the end-to-end distance 'operator':

$$\vec{R}_N = \vec{r}_1 + \vec{r}_2 + \dots + \vec{r}_{N-1} = \sum_i \vec{r}_i.$$

In order for us to compute the physical quantity we would call end-to-end distance we need to calculate the expectation value of $\vec{R}_N$ with respect to the distribution of end-to-end distances. In the spirit of minimalism we will assume again that space is isotropic, i.e., our system does not preferred a particular direction over others, and infer that the probability distribution for end-to-end distances must be direction agnostic, i.e., $P(\vec{R}_N) = P(|\vec{R}_N|)$. This means that $P(\vec{R}_N)$ is an even function, a function for which $P(|\vec{R}_N|) = P(-|\vec{R}_N|)$, while $\vec{R}_N$ is of course odd (it is negative when $\vec{R}_N$ is negative, and positive when $\vec{R}_N$ is positive). This amounts to the overall integral over all R_N of the product $P(\vec{R}_N)\vec{R}_N$ vanishing, namely,

$$\langle \vec{R}_N \rangle = \int_{-\infty}^{\infty} P(R_N) R_N dR_N = 0.$$

So, we conclude that our expectation is that the polymer will be on average in a configuration in which both ends meet.

In order to understand the behavior of the folded configuration it is also instructive to find the fluctuations about the mean, to compute $\langle \vec{R}_N \vec{R}_N \rangle$, i.e.,

$$\langle \vec{R}_N \vec{R}_N \rangle = \left\langle \sum_i \vec{r}_i \sum_j \vec{r}_j \right\rangle = \sum_{i,j} \langle r_i r_j \rangle = \sum_{i,j} l^2 \langle \cos(\theta_{ij}) \rangle$$

with l the monomer-monomer distance and $\theta_{i,j}$ the angle between the intermonomer vectors $\vec{r}_i$ and $\vec{r}_j$. For a pictorial presentation of these quantities see Figure13.1 B.

Now we need to make a decision how to will treat the correlations between the angles between intermonomer vectors. Treating them slightly differently gives rise to two toy models of polymers that we will discuss. In the freely joint chain (FJC) model we make a drastic simplification and assume that all angles no matter how close or how far they are from each other in sequence are perfectly uncorrelated. The ideal chain introduces some correlation in order capture some degree of rigidity in the polymer backbone. We will start from FJC.

Freely joint chain (FJC)

In the freely joint chain model all correlations between orientations of monomer-monomer units are neglected capturing the behavior of a perfectly flexible chain. The $\langle \cos(\theta_{i,j}) \rangle = 0$ for $i \neq j$ and $\langle \cos(\theta_{i,j}) \rangle = 1$ for $i = j$. In this model it is extremely easy to calculate the $\langle \vec{R}_N^2 \rangle$ it is simply

$$\langle \vec{R}_N^2 \rangle = \sum_{i,j} l^2 \langle \cos(\theta_{ij}) \rangle = l^2 N$$

We see that the end to end distance fluctuations grow linearly with the length of the polymer N , the longer the polymer, the farther ends will tend to move away from each other as they are fluctuating in solution.

Ideal chain

We may ourselves whether the fact that most polymer have some rigidity to their backbones and they are not as floppy as assumed in the FJC model may change this result. In this model we will assume that $\langle \cos(\theta_{ij}) \rangle \rightarrow 0$ when $|i - j| \gg 1$ but $\langle \cos(\theta_{i,j}) \rangle \neq 0$ when i and j are close to each other in sequence space. We will define a new quantity called the Flory characteristic ratio

$$C_N = \frac{1}{N} \sum_i C'_i$$

where $C'_i = \sum_j \langle \cos(\theta_{ij}) \rangle$. Using this quantity we may re-write the end-to-end fluctuations as

$$\langle \vec{R}_N^2 \rangle = \sum_{i,j} l^2 \langle \cos(\theta_{ij}) \rangle = l^2 N C_N$$

and we observe again that

$$\langle \vec{R}_N^2 \rangle \sim N.$$

So including some degree of local rigidity did not change the overall result, and we may say that FJC and the ideal chain models form a universality class with respect to scaling of the end-to-end fluctuations with the length of the polymer.

13.2 Entropic origin of rubber elasticity

So far we have managed to avoid deriving the probability distribution for the end-to-end distance in ideal polymers. We will now derive it by imagining that a conformation of a polymer in space is a path of a random walker. We will start with a random walk in 1 dimension (1D) and then generalize to 3D in the very end. In 1D an unbiased random walk starts at the origin and the random walker has an equal probability of making a step to the right (+) as to the left (-), i.e., $p_- = p_+ = 0.5$. For a polymer of length N we will consider walks in which the random walker makes the total of N steps. n_+ of this steps will be made to the right, and n_- of these steps will be to the left, with the total $N = n_- + n_+$. If our origin corresponds to the first monomer of the polymer, then end-to-end distance is captured by how far the random walker gets from the origin after N steps. We may define the end-to-end distance then as $x = (n_+ - n_-)l$ where l is the monomer-monomer distance, or the length of the random walker's steps. Then in terms of x we may express n_+ as $n_+ = \frac{x}{2l} + \frac{N}{2}$.

In order to compute the probability of the random walker to stop x steps away from the origin after N steps we need to estimate the number of paths of length N in which n_+ were made in the positive direction. The number of such paths, $\Omega(n_+, N)$ is quantified using a binomial coefficient, i.e., it is equivalent to the number of ways to choose n_+ objects from a stack of N objects at random and so

$$\Omega(n_+, N) = \frac{N!}{n_+!(N - n_+)!}.$$

To get the probability of such paths we need to divide their number by the total number of paths of length N , so

$$P(n_+, N) = \Omega(n_+, N)/2^N = \frac{1}{2^N} \frac{N!}{n_+!(N - n_+)!}.$$

We may re-write this expression in terms of x to get the probability distribution for the end-to-end distances

$$P(x, N) = \frac{1}{2^N} \frac{N!}{\left(\frac{x}{2l} + \frac{N}{2}\right)! \left(\frac{N}{2} - \frac{x}{2l}\right)!}.$$

Now, working with factorials is tricky, but luckily we are concerned with thermodynamic systems for which N is very large, $N \rightarrow \infty$. In this case we may simplify this expression by following four simple (even if algebraically tedious) steps and obtain the distribution in for the 1D case

$$P(x, N) = \frac{1}{\sqrt{2\pi N} l^2} e^{-\frac{x^2}{2Nl^2}}.$$

In order to generalize to 3D we will assume that the three dimensions are independent and each gets a third of the polymer's total length or $N_x = N_y = N_z = N/3$. We may then write down the 3D distribution as product of 1D distributions:

$$P_{3D}(x, y, z, N) = P_{3D}(r, N) = \left(\frac{3}{2\pi N l^2}\right)^{3/2} e^{-\frac{3r^2}{2Nl^2}}$$

with $r = (x, y, z)$ and N is replaced with $N/3$.

Entropic restoring force

Now is a good time to ask ourselves what can we expect if we attempt to pull the ends of the polymer apart, to stretch it? Will it prefer to stay in the new configuration or will it bounce back to the coiled up conformation? In case of rubber we know what will happen - rubber polymers bounce back. Interestingly, rubber polymers are known to have very weak bonds between polymers and so the restoring force that we clearly observe when we stretch rubber cannot be due to chemical bonding. What does give rise to the restoring force? We will now show that the force comes from polymer's preference for disorder, the same preference for disorder that is captured by the second law of thermodynamics, i.e., it is the Entropy that pulls your fingers when you stretch a rubber band. Let's see how this deep insight emerges from our simple equations.

The entropy of our polymer is given by $S(r, N) = k_B \ln \Omega(r, N)$ where $\Omega(r, N)$ is the number of possible microstates. The probability of distribution of end-to-end distances r in 3D for a polymer of length N is of course

$$P_{3D}(r, N) = \frac{\Omega(r, N)}{\int \Omega(r, N) dr} = \left(\frac{3}{2\pi N l^2}\right)^{3/2} e^{-\frac{3r^2}{2Nl^2}}$$

and we may re-write the entropy in terms of this distribution like so

$$S(r, N) = k_B \ln P_{3D}(r, N) + k_B \ln \int \Omega(r, N) dr.$$

Plugging in the expression for P_{3D} and simplifying we get

$$S(r, N) = -\frac{3}{2} k_B \frac{r^2}{N l^2} + \frac{3}{2} k_B \ln \frac{3}{2\pi N l^2} + k_B \ln \int \Omega(r, N) dr = -\frac{3}{2} k_B \frac{r^2}{N l^2} + S(0, N).$$

Only the first term depends on the distance r - this is the only term that matters to us and I have collected and relabeled the last two terms as $S(0, N)$. Forces are defined as gradients of potential energy with respect to position coordinates and we need to construct the free energy before we can explore the forces that arise due to stretching of the polymers. The free energy is constructed from thermodynamic definition

$$F(r, N) = U - S(r, N)T \approx 0 + \frac{3}{2}k_B \frac{r^2}{Nl^2}T + S(0, N)T = \frac{3}{2}k_B \frac{r^2}{Nl^2}T + F(0, N).$$

In the process of simplifying the expressions we have neglected the term U because in the special case of rubber we do not expect the internal energy to be affected by changes in the coordinate r and we do not expect it to play a role.

The force due to change in end-to-end vector r is given by

$$f = -\frac{\partial F}{\partial r} = -\frac{3k_B T}{Nl^2}r$$

and we can clearly see that as with increasing r there will be a force acting in the opposite direction. This force is called a restoring force, a force that acts to restore the original configuration of the system and in this case, for rubber, the force is entirely due to entropic contributions to free energy.

Chapter 14

Molecular Dynamics Simulations (MD)

NOTE: incomplete notes

Please, be referred to the following sections in the Tuckerman textbook:

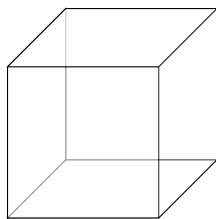
1. The basics of classical mechanics: 1.1-1.3, 1.6, the harmonic polymer is in 1.7
2. Computational molecular dynamics (MD) introduction 3.7, Verlet and Velocity Verlet algorithms: 3.8.1-3.8.2
3. Brownian motion: 15.3.1

14.1 Justification for use of classical mechanics to simulate molecules

This section is focused on justifying the use of classical mechanics to simulate molecular dynamics.

Equipartition of energy and the thermal wavelength of quantum particles

Particle in a box



A box of length, width and height L .

Recall from Quantum mechanics the particle in a box model. The energy is given as

$$E_n = \frac{\hbar^2}{8mL^2}n^2 = \frac{\hbar^2}{8mL^2}(n_x^2 + n_y^2 + n_z^2)$$

for quantum numbers 'n'.

$$Z = \sum_{n_x}^{\infty} \sum_{n_y}^{\infty} \sum_{n_z}^{\infty} e^{-\beta E_{n_x, n_y, n_z}} = \sum_{n_x}^{\infty} \sum_{n_y}^{\infty} \sum_{n_z}^{\infty} e^{-\beta \frac{\hbar^2}{8mL^2} n_x^2} e^{-\beta \frac{\hbar^2}{8mL^2} n_y^2} e^{-\beta \frac{\hbar^2}{8mL^2} n_z^2} = \left(\sum_{n_x}^{\infty} e^{-\beta \frac{\hbar^2}{8mL^2} n_x^2} \right)^3$$

14.1.1 De-Broglie Thermal Wavelength

$$\Lambda = \frac{\hbar}{\sqrt{2\pi m k_B T}} \Rightarrow -\beta \frac{\hbar^2}{8mL^2} n_x^2 = -\frac{\pi}{4} \left(\frac{\Lambda}{L} \right)^2 n_x^2 \Rightarrow Z = \left(\sum_{n_x}^{\infty} e^{-\frac{\pi}{4} \left(\frac{\Lambda}{L} \right)^2 n_x^2} \right)^3$$

The system can be considered slow-varying when $\frac{\Lambda}{L} \ll 1$ and the sum can be treated as an integral, so that

$$\sum_{n_x=1}^{\infty} e^{-\frac{\pi}{4} \left(\frac{\Lambda}{L} \right)^2 n_x^2} \approx \int dx e^{-\frac{\pi}{4} \left(\frac{\Lambda}{L} \right)^2 n_x^2} = \frac{L}{\Lambda}$$

since we recognize that this is a Gaussian integral. Then

$$\Rightarrow Z = \left(\frac{L}{\Lambda} \right)^3 = \frac{V}{\Lambda^3}$$

for one particle. For N particles,

$$Z_N = \frac{1}{N!} (Z^N) = \frac{1}{N!} \left(\frac{V}{\Lambda^3} \right)^N$$

This equation can be broken up by taking logarithms and applying logarithm rules,

$$\log(Z_N) = -\log(N!) + N \log(N) \frac{V}{\Lambda^3}$$

Note that the first term does not depend on T but the second term does. Then

$$\begin{aligned} \bar{E} &= -\frac{\partial \log(Z)}{\partial \beta} = -N \frac{\partial}{\partial \beta} \log \left(\frac{1}{\Lambda^3} \right) = -3N \frac{\partial}{\partial \beta} \frac{\log(\sqrt{2\pi m k_B T})}{h} \\ &\Rightarrow -3N \frac{\partial}{\partial \beta} \log(\sqrt{k_B T}) = -3N \frac{\partial}{\partial \beta} \log \left(\frac{1}{\sqrt{\beta}} \right) = -3N \frac{\partial}{\partial \beta} \log \left(\left(\frac{1}{\beta} \right)^{-\frac{1}{2}} \right) \\ &\Rightarrow = \frac{3}{2} \frac{\partial}{\partial \beta} \log(\beta) = \frac{3}{2} N \frac{1}{\beta} = \frac{3k_B T}{2} N \end{aligned}$$

leading to one of the most important results of this section.

$$\bar{E} = \frac{3k_B T}{2} N \tag{14.1}$$

Modeling Liquids:

When looking at models of liquids, usually we don't consider quantum-mechanical effects. Even for small systems with only 100,000 molecules considering quantum effects is insane.

Usually, quantum mechanical effects are excluded if the typical distance (which we denote ' σ ') between particles is greater than the De-Broglie thermal wavelength, i.e. $\Lambda \ll \sigma$. For numerical constraints, usually $\sigma \approx 5\text{\AA}$ and $\Lambda \approx 0.2\text{\AA}$ which is a factor of approximately 25, so classical molecular dynamics (MD) is ok to use.

14.2 Analytical solutions to molecular dynamics trajectories: the case of a harmonic polymer

It is possible to obtain the trajectories of simple molecular systems by solving the equations of motion analytically. In the general case analytical solutions become too complex and impractical but here we will take a look at what it means to obtain the time dependence of the positions and momenta/velocities for the case of a polymer in which monomers are held together by harmonic potentials. The derivation is given in Tuckerman 1.7 and was done in class as well, I will not repeat it here because we followed Tuckerman very closely.

14.3 Computational Molecular Dynamics

In classical computational molecular dynamics simulations we treat particles as masses on springs. The energy function $E_{system}(t) = E(\text{positions and momenta at time } t) = T(\text{momenta}, \vec{p}) + U(\text{positions}, \vec{r})$ is constructed from the kinetic energy terms T and the potential energy U and it determines the dynamics of our system. The part of the energy function that gives rise to the forces that drive dynamics is of course the potential energy part U and this is the part that includes the molecule-specific energy terms - it includes all the bonding and non-bonding (NB) interactions between the atoms in our system. We may obtain this energy from first principles by solving the time-independent Schrodinger equation, however this is very expensive and instead we fit the quantum mechanical energy function onto a simpler function by obtaining parameters through various machine learning procedures such that our function gives a reasonable approximation true energy. For example an energy function could be given by

$$U(\vec{r}) = \sum_{bonds} \frac{1}{2} k_{bond} (r_{bond} - r_0)^2 + \sum \frac{1}{2} k_{bend} (\theta_{bend} - \theta_0)^2 + \sum_{tors, n} A_{tors} [1 + \cos C_n \phi_{tors} + \delta_n] + \\ + \sum_{i, j \in NB} [4\epsilon_{ij} ((\frac{\sigma_{ij}}{r_{ij}})^{12} - (\frac{\sigma_{ij}}{r_{ij}})^6) + \frac{q_i q_j}{r_{ij}}]$$

where the various terms capture the energy contributions of the different types of interactions (bonded and non-bonded) within our molecular system. The positions of atoms in our system may vary but the coefficients $k_{bond/bend}$, r_0 , θ_0 , C_n , δ_n , ϵ_{ij} , σ_{ij} , etc.. are chosen in a way that makes this energy function be a good approximation to the true quantum mechanical energy function for a given molecule. The combination of the choice of the functional form of U and the set of parameters that represents a molecule is called a 'force-field'. There are many

off the shelf force-fields, that had been developed over the decades, available to choose from, each with its own advantages and draw-backs. Here we will not discuss how force-fields are developed but we will assume that we are dealing with a molecule that is described by some complicated potential energy function. Our goal is to obtain dynamics and we will take the energy function for granted.

Solving the Hamilton or Newton's equations of motion will generate trajectories in which the energy is conserved. Often we need to introduce temperature into our simulations, and as we relax the energy constraint - our dynamics becomes dissipative. There are many ways to simulate a dissipative thermal NVT ensemble but the most popular and intuitive approach is through augmenting the energy conserving equations of motion with dissipative forces such as friction and a random thermal driving force. The resulting stochastic equation of motion is called the Langevin equation and it can be solved analytically for simple cases such as Brownian motion (diffusion in the absence of deterministic forces). In general, however, numerical solutions are needed.

In this section, we will discuss numerical solutions to molecular dynamics given arbitrarily complex potential energy functions which are typically given by commonly used force-fields derived from quantum mechanical calculation of the energies as a function of molecular coordinates through fitting procedures or machine learning. We will not discuss in detail how the energy functions are obtained for different molecules, but we will assume that there is some complex energy potential that our molecules obey and discuss the dynamics algorithms that are used to obtain the trajectories in the phase space in two cases: 1. when energy is conserved and 2. when temperature is conserved.

For more details see Ch. 1 and Ch. 3 in Tuckerman and/or class notes.

14.3.1 Integrating the equations of motion numerically

Given a complex potential energy function U we will develop a numerical scheme called the Verlet algorithm to propagate the positions and momenta/velocities of our particles numerically one time-step at a time.

Verlet algorithm

See section 3.8.1 in Tuckerman.

Velocity Verlet algorithm

See section 3.8.2 in Tuckerman.

14.3.2 Langevin equation and Brownian motion

Langevin equation

The *Langevin Equation* describes diffusive motion of a particle or particles subject to potential energy U and in contact with a thermal 'bath' at temperature T via 'friction'

dissipative/non-conservative forces introduced through the parameter γ . Equations of motion for a single particle with position coordinate r , velocity v , and mass m are the familiar Newton's equations augmented with a dissipation/driving terms:

$$\dot{r} = v$$

$$\dot{v} = -\frac{\gamma v}{m} + \frac{F_r(t)}{m} - \frac{1}{m} \frac{\partial U(r)}{\partial r}$$

where $m, v, F_r(t), U(r)$ are mass, velocity, random thermal force (importantly, the random force is zero on average), and the potential energy respectively. The first term with the "constant" γ is the dissipative *friction* term, the middle term is called the *random driving force* term and the term on the right is the *deterministic force*.

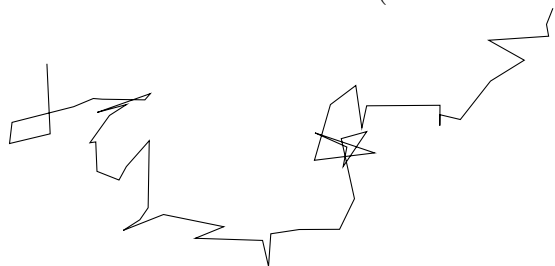
Langevin equation is an equation of motion augmented to include dissipative forces, namely, friction. Friction is a term that depletes acceleration with a force proportional to velocity, hence we add the familiar $-\gamma v$ to the Newton's equations of motion, where γ is the friction coefficient. Dissipating energy, however, is only one 'action' of friction, only one effect. The other effect is the injection of energy into the system through non-directed, thermal, random 'kicks'. This effect of the outside environment will be included in the equations of motion through a random force $F_r(t)$. The presence of a random variable in our differential equations makes them stochastic, and the resulting trajectories are analogous to the random walks we have discussed in the context of polymers. Indeed, because the dissipation of energy and the injection of energy are related to the phenomenon we call 'friction', there exists a relationship between the friction parameter γ and the thermal properties of the random force $F_r(t)$. This relationship is established by the **Fluctuation-Dissipation theorem** which states that the random force needs to vanish on average, i.e., $\langle F_r(t) \rangle = 0$, but the time-correlations of the force obey

$$\langle F_r(t') F_r(t) \rangle = 2m\gamma k_B T \delta(t - t')$$

where δ is the Dirac delta-function. It's easy to see that temperature enters the dynamics through the statistical properties of the random force. This is the term that ensures that our simulations produce thermal trajectories which may be used to calculate expectation values of observable with respect to the NVT ensemble.

Brownian motion

The special case of a freely diffusing particle in the absence of external potentials $U(r) = 0$ is called Brownian motion (see Tuckerman 15.3.1).



An example of a trajectory of a Brownian particle: thermal motion in the absence of deterministic forces $U(\vec{r}) = 0$.

In class we have derived the diffusion coefficient for a Brownian particle in terms of the velocity correlation function. We found it to be

$$D = \int_0^\infty \langle v(0)v(t) \rangle dt = \int_0^\infty \frac{k_B T}{m} e^{-m\gamma t} dt = \frac{k_B T}{\gamma}.$$

For full derivation see Tuckerman section 15.3.

Calculating expectation values of observables

-time average:

$$\langle O \rangle = \lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t O(x(t)) dt$$

-ensemble average:

$$\langle O \rangle = \int_{-\infty}^{\infty} P(x) O(x) dx$$

where O is some observable (position, kinetic energy, etc...) and $P(x)$ is the relevant probability distribution. For example, simulating an N, V, T canonical ensemble:

$$\langle O \rangle = \frac{1}{Z} \int e^{-\beta E(x)} O(x) dx$$

The **ergodic hypothesis** states that there exists equivalence between time averages of dynamical systems and ensemble averages (of many replicas) of static systems making dynamical trajectories useful for calculating the expected values while bypassing the calculation of the problematic $P(x)$ explicitly.

In practice, we simply run dynamics, collect the trajectories, and calculate the averages of quantities of interest with respect to this dynamical data: snapshots from our simulations in effect replace the static ensemble and we approximate our observables as

$$\langle O \rangle \approx \frac{1}{M} \sum_{i=1}^M O_i$$

where M is the number of time-steps in our trajectory and O_i is the value of the observable of interest at each time-step. Longer and more accurate trajectories will produce better approximations to the real expected values of observable O .

Solving the Langevin Equation on a computer

With the notable exception of Brownian motion (see Tuckerman 15.3.1 and our class notes), the Langevin equation is difficult to solve analytically for an arbitrarily complicated potential $U(\vec{r})$. The equations we would like to solve are these:

$$\frac{dr_i}{dt} = v_i$$

$$\frac{d^2 r_i}{dt^2} = -\frac{\gamma}{m_i} v_i(t) + \frac{F_r(t)}{m_i} - \frac{1}{m_i} \frac{\partial U(\vec{r})}{\partial r_i}$$

Here's how we make a single step forward for the i^{th} particle in our simulation trajectory:
At time $t = t_0$ calculate the deterministic forces F_d at position $r_i(t_0)$:

$$F_{d,1} = -\frac{\partial U(\vec{r})}{\partial r_i(t_0)}.$$

Use the deterministic forces $F_{d,1}$ to find the intermediate velocities

$$v_{temp,1} = e^{-\gamma\delta t/2} v_i(t_0) + \sqrt{\frac{k_B T}{m_i}} \sqrt{1 - e^{-\gamma\delta t}} R^{Gaussian}(0, 1) + \frac{1}{2} \frac{F_{d,1}}{m_i} \delta t.$$

where $R^{Gaussian}(0, 1)$ is a random real number sampled from a Gaussian distribution in the interval between 0 and 1.

Now, update the positions using intermediate velocities $v_{temp,1}$:

$$r_i(t_0 + \delta t = t_1) = r_i(t_0) + v_{temp,1} \delta t.$$

Update the deterministic forces again, this time at the new position $r_i(t_1)$:

$$F_{d,2} = -\frac{\partial U(r_i(\vec{t}_1))}{\partial r_i},$$

and use them to update the intermediate velocities once more:

$$v_{temp,2} = v_{temp,1} + \frac{1}{2} \delta t \frac{F_{d,2}}{m_i}.$$

Finally, we construct the velocities at time $t_1 = t_0 + \delta t$, $v(t_1)$:

$$v_i(t_1) = e^{-\frac{\gamma\delta t}{2}} v_{temp,2} + \sqrt{\frac{k_B T}{m}} \sqrt{1 - e^{-\gamma\delta t}} R^{Gaussian}(0, 1)$$

where $R^{Gaussian}(0, 1)$ is a random real number sampled from a Gaussian distribution in the interval between 0 and 1. To make the next time-step simply repeat these steps using $\vec{r}(t_1)$ and $\vec{v}(t_1)$ as your starting points. Repeat as many times as needed to generate a trajectory that is sufficiently long to give your access to the phenomena you study and to obtain good estimates for expectation values of observables of interest.